

FACULTY OF
COMPUTER SCIENCE

Otto-von-Guericke-University Magdeburg

Faculty of Computer Science

Department of Databases and Software Engineering

Comparative Analysis of Multimodal Emotion Detection Approaches

Bachelor Thesis

Author:

Lukas Eichel

Examiner:

Prof. Dr. rer. nat. habil. Gunter Saake

2nd Examiner and Supervisor:

Dr.-Ing. David Broneske

Magdeburg, 13.09.2024

Abstract

Emotion are an important aspect in general human understanding as they lead to better understanding of conversation and other daily behavior. Many psychologist defined rules for emotion detection like different movements of muscles in faces or significant changes in the voice while speaking. Therefore many patterns were observed and aligned with a specific set of emotions.

Deep learning is a well researched topic, known for many kinds of pattern recognition. Nowadays transformer models like ChatGPT are getting more and more popularity and they are said to be all-purpose tools for many tasks. This thesis is implementing state-of-the art transformers for emotion detection in text, audio and video. These were not coverd by research in emotion detection with artificial intelligence yet. Furthermore it compares these with traditional convolution models and gets in touch with multimodal recognition, where the results are combined together through their labels.

The results are showing, that transformer architectures are able to recognize emotions. Classification in videos showed by far very promising results. However this thesis showed that emotions are context specific and the state-of-the art models nor the convolution models are able to recognize these beyond their dataset.

Contents

1. Introduction **1**
- 1.1. Motivation . . . 1
- 1.2. Structure . . . 2

2. Background **3**
- 2.1. Emotion . . . 3
  - 2.1.1. Emotion Classification . . . 4
  - 2.1.2. Emotion in Text . . . 5
  - 2.1.3. Emotion in Speech . . . 6
  - 2.1.4. Emotion in Facial Expression . . . 7
- 2.2. Classification Approaches . . . 8
  - 2.2.1. Classification in Text . . . 9
  - 2.2.2. Classification in Audio . . . 14
  - 2.2.3. Classification in Images and Videos . . . 17
- 2.3. Datasets . . . 20

3. Related Work **23**
- 3.1. Audio and Video Based . . . 23
- 3.2. Audio, Video and Text Based . . . 24

4. Design and Implementation **27**
- 4.1. Unimodal Text Classification . . . 27
  - 4.1.1. Preprocessing and Prompt Engineering for LLM . . . 29
- 4.2. Unimodal Audio Classification . . . 32
- 4.3. Unimodal Video Classification . . . 36
- 4.4. Multimodal Classification . . . 40

5. Evaluation **44**
- 5.1. Setup . . . 44
  - 5.1.1. Accuracy Metrics . . . 44
  - 5.1.2. Experiments . . . 45
- 5.2. Results . . . 46

6. Conclusion **58**
- 6.1. Summary . . . 58
- 6.2. Limitations . . . 59
- 6.3. Future Work . . . 59

Bibliography **62**

Chapter 1.

Introduction

1.1. Motivation

Both artifical intelligence and deep learning architectures are currently undergoing a rapid development with ever-increasing optimization efforts. With the introduction of transformer models user-oriented applications to generate text, audio and videos or to identify everday sequences which are featuring trainable logic. Furthermore, they have been declared as all-rounder tools, capeable of slowly rendering the first jobs obsolete. This begs the question if they are really able to work as a multi-tool or if they just profit from the large range of training data and if there will be some kind of bias due to coarse data with less depth and no focus on a special task like emotion detection.

From an early stage of emotion research, psychological approaches have shown fixed behavioral patterns when showing emotions and sentiments including face and voice actions regarding semantic and syntax. Basic math and machine learning algorithms were able to classify some information and identify emotion. However neural networks are bringing more performance, better results in less time. The introduced transformers are infamous for their contextual understanding as well as very good and consistent outcomes from the given tasks.

This leads to the question if already developed applications like ChatGPT or general approaches based on deep learning are able to recognize emotion from video, audio and text in the context of interviews with a static view-angle and real-life excerpts. The goal of this thesis is to compare state of the art models and architectures mostly focused on transformers which are present in every modal. Further multimodal emotion detection is a topic to analyze different feature fusion models and wether the different labels from the different architectures might lead to a more accurate results than a single modal outcome. This topic covers basic voting strategies up to further more complex architectures which are using the logits, kernels or other raw output from the models. A comparison can be done to show if the effort is necessary for an accurate result or if given basic methods are nearly as good or even better. These points sequels into an assumption that bimodal emotion classifcation can be more efficient and asymptotically accurate. Generally looking at unimodal classification which might show that one of the modals might have a small impact on the results but not an crucial one. Intuitively this can be emotion classification with features from text due to the impact of irony or other vocal features bringing an ambigious meaning for sentences or words related to their emotion label. The opportunity to ignore one or more modals offers several optimization opportunities

as for example one large architecture less is needed or the chosen feature combination method has a reduction in complexity and computation time.

In general, emotion detection introduces an interesting field of opportunity for research regarding machine-based simulation of human behavior and understanding of interactions related to interviews or conversations. Additionally, after learning from the features, most generative models are able to reproduce emotion alongside the trained information giving perspective in robotics or other forms of real world imitations. Current research is still focused on the general, non-transformer based models like convolution, reccurent networks or even simpler machine learning approaches. Due to limitation from copyright or closed-source projects some architectures have been reproduced but not trained on the original datasets, possibly leading to different accuracys in comparison to the usage of the original model with the pretrained weights.

1.2. Structure

This work starts by showing and discussing the fundamental, psychological backgrounds in Chapter 2 for the term emotion and which features are necessary to identify them. Furthermore, the classification approaches in machine and deep learning are presented showing their unsupervised and supervised methods for training and feature extraction. At the end of this chapter the datasets for the further implementation parts are introduced. Related works in Chapter 3 are presented to display current research on multi-modal detection and their different methods and learning proceedings. Continuing with the design and implementation of the different supervised models, Chapter 4 presents self-trained and also pretrained models and how to expand as well as finetune their knowledge. In addition the architectures to fuse the information for multi-model emotion detection are shown and different fusion methods are discussed including some kinds of basic scoring and more advanced label combinations. These accomplishments are evaluated in Chapter 5 introducting measurement criteria for reliable and comparable results. Finally, this thesis is summarized and a comperative overview is given to contrast some outcomes from related work concluding with a look on possible future work in emotion detection.

Chapter 2.

Background

This chapter will give an introduction and overview on different approaches to classify types of emotions with respect to their advantages and disadvantages. Furthermore, it will explain the approaches and assumptions to recognize emotion in text, video and audio. Based on this, common implementation techniques for supervised and unsupervised approaches are presented with a focus on convolution and transformer models as they are the leading architectures in deep learning. At the end, datasets are introduced which represent the foundation for the later implementation and evaluation of the models.

2.1. Emotion

There is no unambigious definition for the term ‘emotion’. There are many different emotions, with different levels of intensity. Considering the question of what an emotion is and where it comes from, William James discussed two different theories about their origins. Either there are separate areas in the cerebrum or else they correspond to motory and sensory processes [1]. If they originate from the separate areas, the research would be much easier. They could obsere these regions while testing different actions and influences on the test subjects. The results should be quite similar between most people. However, it is most likely the case, that emotions correspond to these motory and sensory processes. They will differ from human to human in intensity and execution, so researchers have to study the small similarities and, additionally, the possible patterns and variations from the shown processes. In this case these factors and all brain activities must be captured to start more research in this topic. Many works in psychology add further aspects to the theory. In order to provide a framework, the following chapters will focus just on the given emotion and how to recognize it in written text, speech and facial expressions. These are the most common expressive features captured by emotion theorists, but there are many more like gestures or when looking at the medical field. Blood pressure and other factors might have an approach for determining the leading emotion as well.

On the other hand there have been first approaches to simulate sentiments with computers or robot-like animatronics. Learned patterns can be applied from a simple command to emotion features which are currently still very overacted.

Figure 2.1 presents aforementioned information such as motory, sensory processes, their combination and the differentiation from semantic and syntatic context. They are categorized in high-level and low level information to process an emotion.

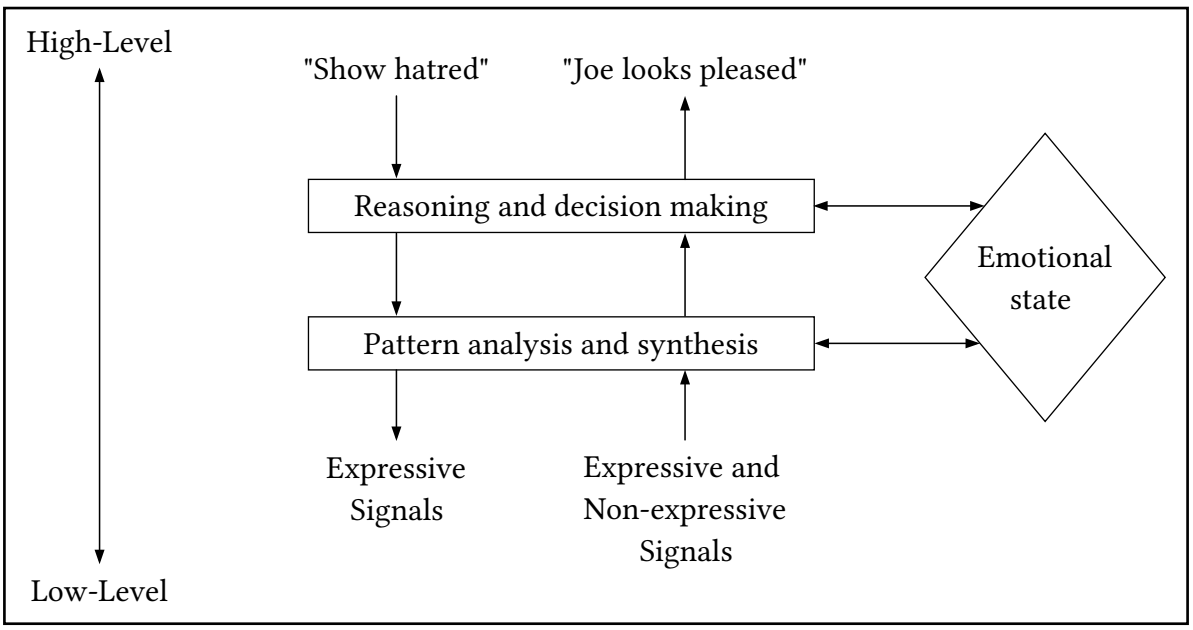


Figure 2.1.: Information flows from both high to low to high in a system that can express, recognize and "have" emotions [2].

High-level information is defined by the semantic of a situation which can be drawn from audiovisual processes, for example, the command "Show hatred". To show hate, a certain emotional state has to be reached and must be reasoned by the situation. As a conclusion low-level information, for instance, face expressions, gestures or other syntactic articulations can be observed from the person or system wich executes the given instruction. Before, these patterns are analyzed or learned and synthesized by the process.

On the other hand, these low-level expressive information can be combined to get the semantic of the emotional state. In addition there can be non-expressive data as valid input. It describes the passive influence on a persons state like the weather or the environmental conditions. Most datasets will not cover this type of information.

The mid-level builds patterns for the given input and combines different low-level signals to synthesize better reasoning for a high-level output. For example a slightly faster speech rate and a tense articulation can be clustered together to get classified as a pleased look.

2.1.1. Emotion Classification

Several apporaches to classify the semantic high-level features have been developed over time. The most popular classification is the basic-emotion framework by Paul Ekman [3] describing six different emotions: anger, disgust, fear, surprise, sadness and happiness. These arose from various facial expressions, which Ekman categorized to the corresponding emotion.

Over time, diverse frameworks were built on this foundation to introduce for each basic emotion multiple detailled subcategories. A common model is the Plutchik wheel of emotion, which is conventionally used to classify emotion in text. It offers eight types of basic emotion: joy, trust, fear, surprise and their opposites sadness, disgust, anger and anticipation [4].

Visible in Figure 2.2, each emotion has multiple levels of intensity. For example, the enhanced level for disgust is loathing, a more moderate level is boredom. In total there are 24 different emotions. A variety of sentiments and their intensities lead to a proper illustration of the real value.

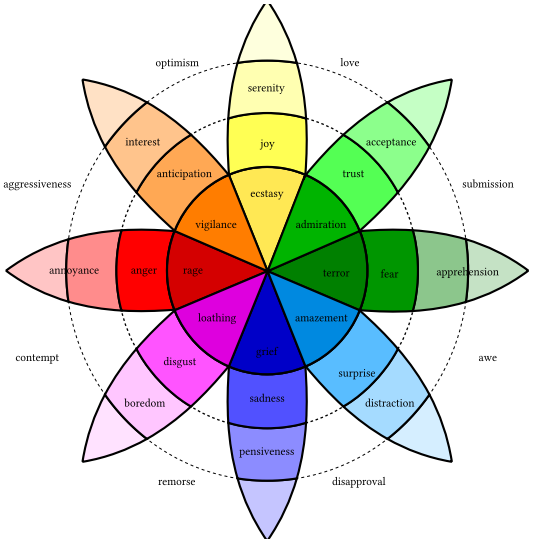


Figure 2.2.: Plutchik Wheel of Emotion [5].

Although there will be a loss of detail, the content of this thesis will focus on the basic emotion as defined by Paul Ekman. The wheel of emotion from Plutchik is suitable for emotion detection in text but most datasets in facial emotion recognition are focused on Ekmans emotion theory. As shown later in this thesis, many multimodal datasets are based on the work from Ekman. It is harder to distinguish all emotion from Plutchik in audio processing and face recognition.

2.1.2. Emotion in Text

The most common approach for classifying emotion from natural language text is to classify the semantic content of these sentences. In order to make this possible, two syntactical assumptions are needed.

1. There is a function that is able to map words and sentences to their corresponding emotion.
2. Words and sentences do not change their meaning over time.

The first assumption can be approached by training a deep neural network with a suitable dataset or a clustering algorithm that measures the similarity between words and sentences to match their corresponding emotion label. The function must be surjective, so a token will not be able to be labeled with more than one emotion at the time. The second assumption is important for a reliable result in training but it is also the flaw of emotion detection in text. For example the sentence “I’m glad to be here”, expressions like irony, emphasis in the voice or gestures can change the proper meaning of the sentence. It could be interpreted as happiness or as disgust. To take this into consideration further observations of other expressive and non-expressive signals are needed.

2.1.3. Emotion in Speech

Emotion in audio is in contrast to the classification in text mainly focused on the syntax from the recording. The actual signal input is represented as a waveform and sinoid functions. A single wave is defined by

$$g(t) := A \sin(2\pi(\omega t - \varphi)) \tag{1}$$

A complete waveform as presented in Figure 2.3 and Figure 2.4 consists of many different sinoid functions which are added and concatenated together [6, pp. 19-22].

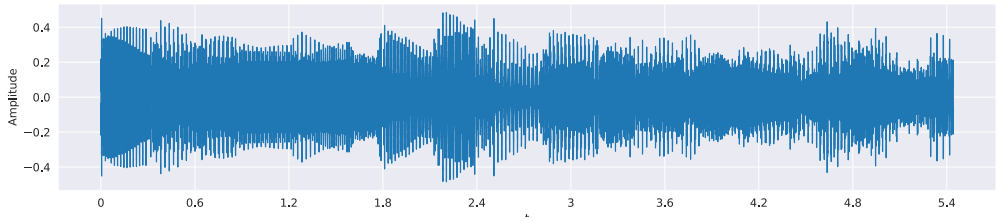


Figure 2.3.: Waveform of the first seconds of an audio recording

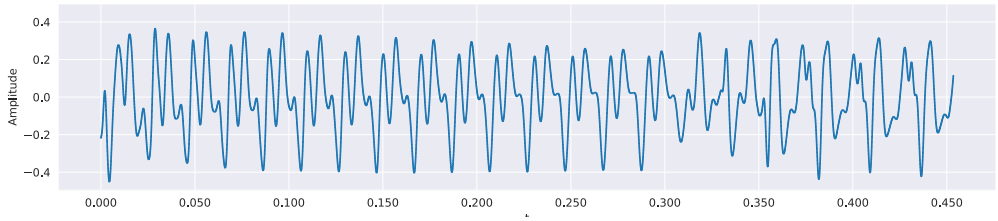


Figure 2.4.: Enlargement of the first half second from Figure 2.3

Frequency (ω) measures the pitch from the audio and especially from the voice. As shown in Table 2.1 multiple features can be extracted from the frequency or to be more precise the pitch. From a measured neutral foundation, changes like an average pitch, the range or other changes can be observed. For example a person who speaks in a much higher pitch — with a wide variablity in the range and abrupt frequency changes on stressed syllables — compared to the average person, will most likely feel anger.

Amplitude (A) shows the intensity and the loudness of the given audio. Furthermore it is also an important feature and easy to extract from speech with the help of an existing neutral foundation.

Phase (φ) shifts $g(t)$ along the x-dimension. This parameter is not important in the feature measurement but has a significant role in audio decomposition which is described later.

Extracting features like speech rate, articulation or voice quality requires more observation. Speech rate can be measured with words per second, sentences per minute or other similar metrics. Normally when looking at waveforms, the intermissions between sentences or words can be retrieved. A higher rate is equivalent to more observed gaps in the same time of a normal speech speed. Articulation and voice quaility is hard to track without semantic knowl-

edge. Therefore a combination of the already presented features is needed which can be done by pattern matching and comparison to the neutral foundation.

Emotion / Feature	Fear	Anger	Sadness	Happiness	Disgust
Speech Rate	much faster	slightly faster	slightly slower	faster or slower	very much slower
Pitch Average	very much higher	very much higher	slightly lower	much higher	very much lower
Pitch Range	much wider	much wider	slightly narrower	much wider	slightly wider
Intensity	normal	higher	lower	higher	lower
Voice Quality	irregular voicing	breathy chest tone	resonant	breathy blaring	grubled chest tone
Pitch Changes	normal	abrupt on stressed syllables	downward inflections	smooth upward inflections	wide downward terminal inflections
Articulation	precise	tense	slurring	normal	normal

Table 2.1.: Features of an audio with correlating emotion [2], [7].

2.1.4. Emotion in Facial Expression

Besides his theory about the six basic emotions, Paul Ekman introduced a system to map different face expressions to their relating sentiments by combining changes in different areas of the expression field. This facial action coding system (FACS) is based on these movements which are called action units. There are 28 main actions which can be mapped to the six main emotion. In the course of this thesis, they will be used as the main features.



Neutral Sad Fear Disgust

Figure 2.5.: Examples with their annoations from the CREMA-dataset [8] which show emotions based on facial expression features.

As visible in Figure 2.5, there are distinctive combinations of face features able to represent different emotion. For example, to observe disgust, the typical wrinkling of the nose, the contraction of the eyebrows and a lifting of the upper lip can be perceived. In addition to the movement of the eyebrows, sadness is marked through the downturned corners of the mouth. Thus a face and its expressions can be broken down into seperate regions.

Eyes (upper face): The most action units to recognize emotion within facial expressions are defined by movement near the eyes. Eyebrows are seperated into an inner and outer part for a more detailed classifcation. Furthermore they can be raised or pulled together leading into four different states. Besides eyebrows, eyelids are another meaningful part in that area. They can be tensed, raised by opening the eyes wide and lowered by narrowing the eyes. In total this region plays an important role for every basic emotion by Ekman and a relevant part for most sentiments based on Plutchiks theory.

Nose: This face region only impacts one basic emotion, disgust, shown in Figure 2.5. This is due to the limited range of movement of this part of the head. Thus this section of the face will not differ from the upper part of the face and the lower part described above due to its low occurrence.

Mouth (lower face): The cheeks and the chin are also correlated to this area for the following work in this thesis due to their tight connection to the lips and mouth by means of muscles. This leads to a total of 16 different action units due to the numerous possibilities, e.g. deforming the lips, opening and closing the mouth and the resulting movements in this area.

In his research Ekman defined each action unit and their corresponding emotion in detail which can be looked up in his book about the basic FACS System [9].

Furthermore in 2002 [10] he updated his idea of the facial expression system and integrated more action units adding up to more than 40 possible features in total. This includes tilting and rotating the head and more details on the upper face like winking or blinking with the eyes.

In addition, a closer look into the various intensities of an action are presented and classified into 5 levels: trace (A), slight (B), marked or pronounced (C), severe or extreme (D) and maximum (E). Most common are slight or marked and pronounced, while the higher intensities are visible in films or other acted illustrations. Maximum intenseness is the most rare but can be found in theater or generally live action plays. The different letters are used as an indicator together with the numerated action unit for a better description and overview of all possible combinations.

2.2. Classification Approaches

Over time many approaches to detect and classify emotion have been developed. There are several supervised architectures and unsupervised techniques which represent the current state of the art for emotion detection in text, audio and video.

2.2.1. Classification in Text

To classify emotions in text, multiple preprocessing steps are necessary due to the lack of numerical values. In the following section, some ideas for preprocessing are presented. Furthermore, a look into various classification methods will be taken. A special focus on the supervised methods is applied due to better comparability of already existing and pretrained deep learning models.

Word Embedding and Tokenization

The most important part of all classification approaches is called word embedding. It represents words as real-valued vectors [11]. It should use the syntactic but also the semantic meaning of the words to get a useful vector. This technique is commonly used in natural language processing like information retrieval in text.

A naive approach to generate such an embedding function would be a one-hot vector encoding [12]. It represents every word in a space as a different vector. For example the sentence:

The tree is green.

will be first tokenized into a set of words V.

$$V = \{\textit{the}, \textit{tree}, \textit{is}, \textit{green}\}$$

The size of the set $|V| = 4$. As a result a mapping $w \in V$ to a $|V|$-dimensional vector is needed.

$$A : V \to \mathbb{R}^{|V|}$$

$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$
the	$\begin{pmatrix}1\\0\\0\\0\end{pmatrix}$	tree	$\begin{pmatrix}0\\1\\0\\0\end{pmatrix}$	is	$\begin{pmatrix}0\\0\\1\\0\end{pmatrix}$	green	$\begin{pmatrix}0\\0\\0\\1\end{pmatrix}$

Table 2.2.: Possible one-hot vector mapping for the sentence “The tree is green.”

To measure the similarity the cosine similarity is often used [12], [13]. Similar words or synonyms should have a smaller angle between each other, compared to unsimilar words.

$$\text{sim}(w_1, w_2) = \cos(\theta) = \frac{w_1 \cdot w_2}{\|w_1\| \, \|w_2\|} \tag{2}$$

Because the one-hot vector embeds the input word by word without any context, there will not exists any similarity or dissimilarity between the words. This is also the reason why simple one-hot encoding is less used in practise. A semantic consideration will be more useful.

The Continuous Bag-of-Words Model, shortly CBOW, is capable of taking the context of a word as an input to calculate an embedded vector with emphasis on syntax and semantic. It describes a neural network which mainly consists of three layers: the input layer, a hidden layer and the output layer [14].

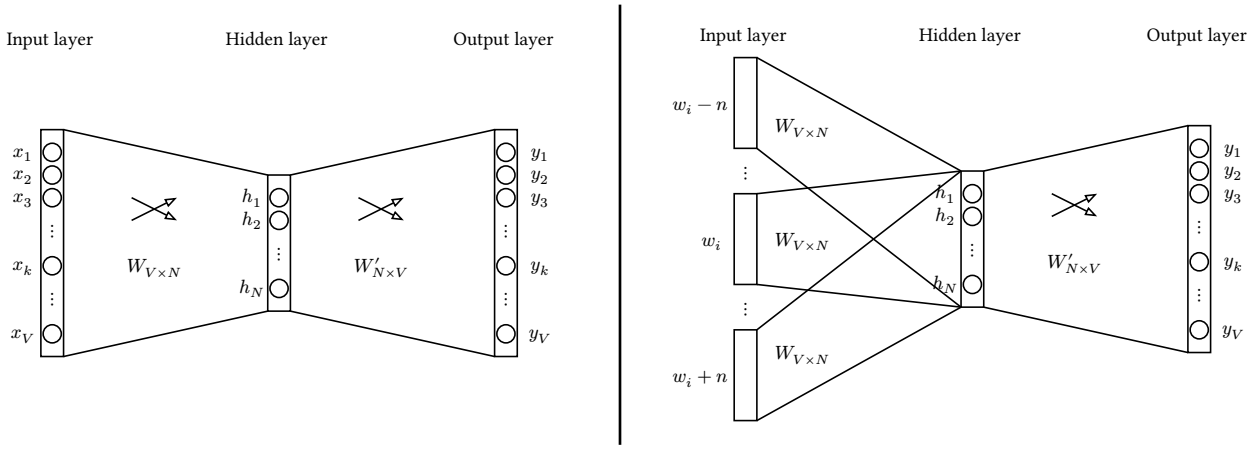


Figure 2.6.: Left: Simple CBOW-architecture for a single word; Right: CBOW-architecture for multiple words. Words have length k and are padded to size V to match the dimension [12].

The architecture shown on the right side of Figure 2.6 takes one one-hot vector $A(w_i)$ as an input expression. To get the context of the inputted term, phrases before and after are needed. Therefore an addition to the architecture is necessary.

With unsupervised learning the model on the right of Figure 2.6 is able to generate a new vector which represents the context of a word. Now, under consideration of the cosine similarity, words with a similar context have a quite smaller angle compared to those without any connection in semantics. In both architectures, the hidden layer acts like an arithmetic operator which combines the context to the outputted one-hot vector.

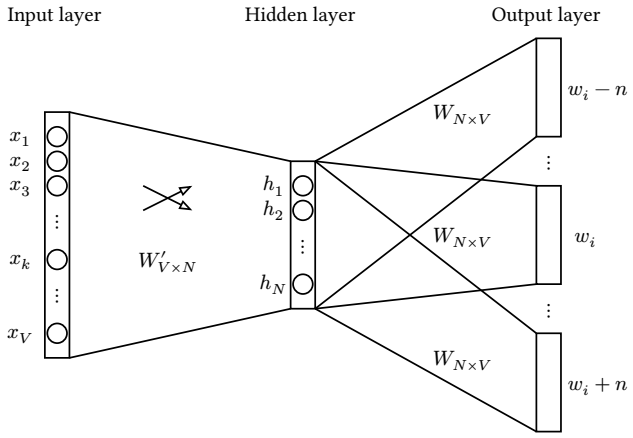


Figure 2.7.: Skip-Gram Model. All words have size V (after padding) [15].

The Skip-Gram Model calculates an embedding in reverse from the idea of the CBOW-Model. As the architecture shows in Figure 2.7 it tries to calculate the surrounding words with a given target term [16]. Therefore a training like the CBOW with contextual phrases is needed before it can work as a semantical decoder. Both architectures are commonly used in different pretrained word embedding networks, like Word2Vec from Google [15], [17].

Fast WordPiece Tokenization is an approach to tokenize sentences and words in an efficient way by splitting words into smaller subwords. It is used in a language understanding transformer model from Google which is called Bert [18].

The basic problem from other tokenization methods like the normal WordPiece Tokenization starts in the vocabulary tokenization in tries. Trie memory is a paradigm which is able to store strings efficiently where each edge of the data structure stores phonemes, the smallest linguistic unit that distinguishes meaning. Traversing a path consisting of multiple edges results in a word. As a result the structure does not have to store redundant parts of words [19].

The tokenization has to search for the longest token at a position. Therefore it starts from the shortest substring and iterates over the input text while trie matching to match the longest prefix which matches a vocabulary token [20]. If the trie matching fails at the next character, the algorithm has to utilize backtracking because the longest vocabulary token may be several characters back. This results in a time complexity of $O(nm)$ where m is the maximum vocabulary token length and n the input length.

Fast WordPiece Tokenization introduces so called failure links and pops. Failure pops are a list of longest-matching prefix tokens of a node in the trie, which can be removed in order until a failure link node is reached. The failure node is able to represent the missing suffix. This whole algorithm called LinMaxMatch has a time complexity of only $O(n)$.

Supervised Classification Approaches

With a given word embedding and tokenization, supervised learning approaches are able to recognize emotion in text. Therefore an annotated dataset is needed, which has a reliable mapping from sentence or word to the corresponding emotion.

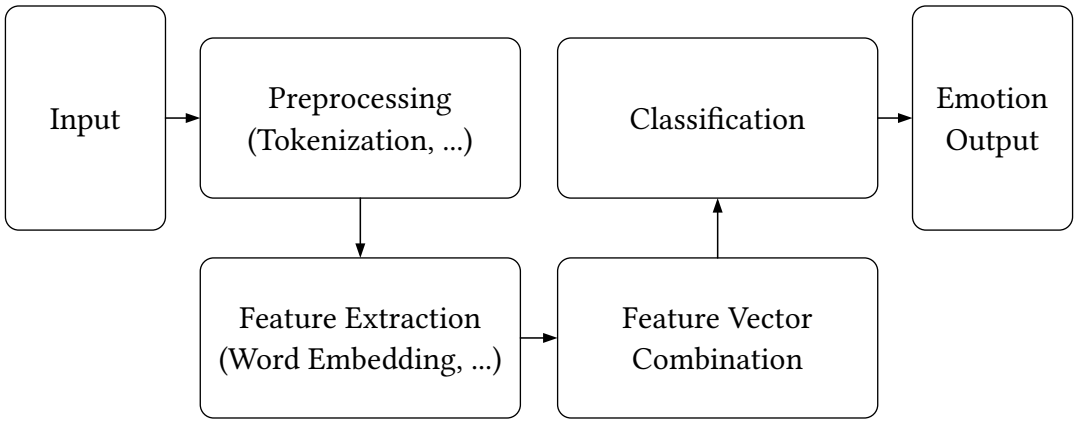


Figure 2.8.: Deep Learning Assisted Semantic Text Analysis

Most models use the architecture paradigm as shown in Figure 2.8. They mostly differ in the step of combining the feature vectors and the type of classification because it offers the biggest room to maneuver. A preprocessing and feature extraction is always necessary for classification tasks. Sometimes a WordNet [21] is integrated to get a better overview about synonyms or to paraphrase words.

Convolutional and Recurrent Neural Networks (CNN/RNN) are an effective approach on sentiment analysis of text [22]. Convolution consists of a kernel-function which downscales the input but keeps the important features. This technique is called Pooling and brings several benefits like less memory usage, less overfitting in training and a faster computation time. Therefore multiple words or sentences of the space V are combined to region matrices. On each region L convolutional filters are applied to learn local n-gram features.

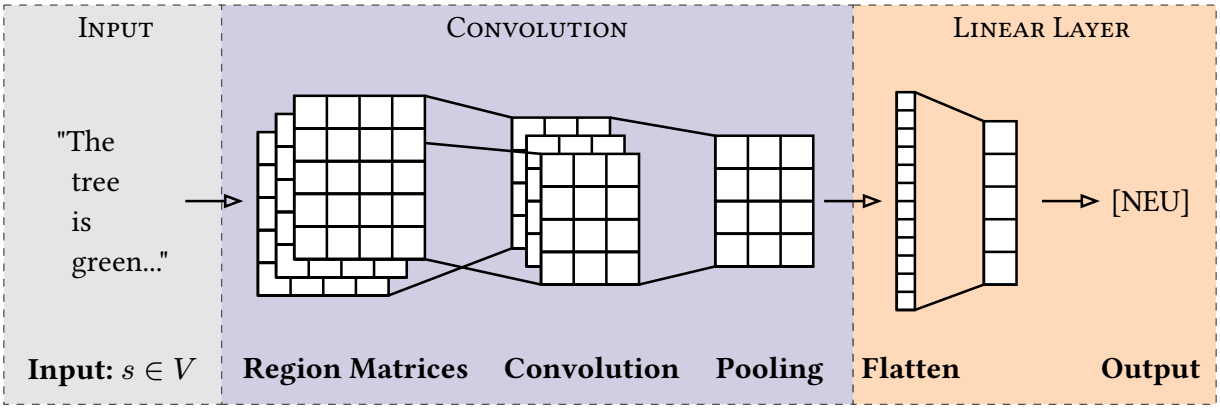


Figure 2.9.: Simple Convolution based Neural Network to predict an emotion from a given subset of sentences

Figure 2.9 shows such a network with the convolutional part. In common architectures multiple convolution and pooling layers are stacked together with an activation function like ReLu in between each block which improves the output by setting constraints to the convolution outputs. The constraint which is affected by ReLu is the prohibition of negative values by the function $f(x) = \max(0, x)$. This sequencing of the convolution blocks leads to a hierarchical decomposition of the given embedded region matrix. As a result multiple features can be found from each block and each convolution kernel which might result in a higher accuracy. At the end an ordinary decoder in form of a simple linear layer with an activation function like SoftMax is utilized to calculate the resulting emotion from the pooled and flattened vector. Normalization such as layer norm or batch norm between each layer can help to ensure better propagation and reduce numerical instability [23].

Further optimizations to fundamental neural network architectures are gating mechanisms from recurrent networks and the bi-directional flow in the model forwarding architectures. They introduce new types of storage cells allowing for long term feature storage [24]. In the context of emotion detection in text, it might store important context from previous sentences and words to differentiate between fill-words like “this”, “with” and emotionally loaded words. Commonly used are long short term memory cells (LSTM) [25] and gated recurrent units (GRU) [26] which both try to prevent gradient vanishing which is a big problem in recurrent networks. It occurs while propagating backwards through time. Due to the load of multiplication steps, the gradient converges to 0. These two architectures differ mostly in their structure than in their functionality. In addition to the GRU cell, an LSTM has two more control gates which gives more possibilities to constrain and rule the memory inside the structure. In both cases they can improve learning features from context and time series.

Transformers are a deep-learning architecture which is based on multi-head attention.

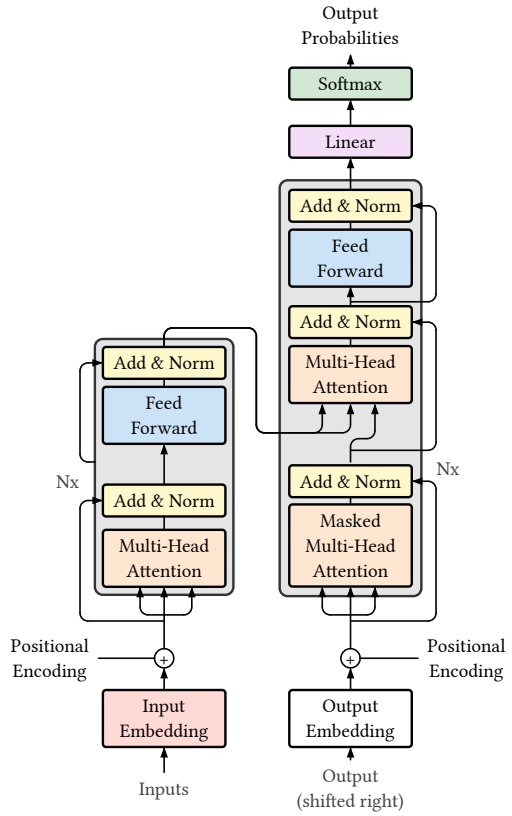


Figure 2.10.: Common transformer model architecture with multi-head attention [27].

The basic idea of attention solves a huge bottleneck in common neural network architectures because fixed input vector sizes may result in a large overhead. In the example of natural language processing, this could be a result from very large sentences. Attention mechanisms try to solve this issue by focusing the input on to specific parts e.g. training the input weights on important words or subsets from the given sentence.

The resulting transformers are currently setting a milestone in natural language processing and providing the basis for ChatGPT from OpenAI [28] or other text understanding models, because of their scalability due to the attention mechanisms and their reliable results. Figure 2.10 shows such a transformer architecture. An input sentence will be fed into the model where it first gets positionally encoded thus the position and the frequency for every word in the sentence is stored.

Now, the input vectors are passing through the first multi-head attention layer to lay the focus on important parts of the input. Nevertheless, the actual input and the weighted part are added and normalized together which results in the original sentence but with tagged highlights due to the attention. To learn the features for the output token, the input traverses through a feed forward network which consists of basic uni-directional layers. Likewise the input and the output are matched together with a residual connection. The same procedure is repeated but with an autoregressive addition of already calculated output. In contrast to the regular multi-head attention, a masking in this process is introduced which sets the weight for unknown or future tokens to negative values thus the model can learn realistic predictions only with already known tokens. The output of the transformer architecture are probabilities for the next state. The already mentioned autoregressive implementation gives the possibilities to predict a long sequence of tokens.

In addition to the approaches with neural network architectures like transformers, other supervised methods like support vector machines, decision trees or naive bayes are used. These are not covered in this thesis as it exceeds the scope of this work and are a possible part for future work.

In summary, supervised methods follow the basic principle to learn features along a given target. Therefore enough training data is needed with the additional annotated labels.

Finally even the differentiation between convolution, recurrent or transformer networks will not make any noticeable changes in measurements if the architectures and models are large enough [29]. In literature this "bitter lesson" [30] is researched step by step and also shown for language models, convolution nets and transformers.

"Scaling Laws for Neural Network" [31] shows the improvement of the loss, when a model has many parameters, a lot of computation time in training and a large dataset. Therefore this thesis will consider the transformer models as a focal point for later evaluation, because there are many large pretrained language models available which are able to recognize emotion and are trained and developed to satisfy almost all three factors. Language modeling performance improves smoothly as the model size, datasetset size, and amount of computation used for training is increased. For optimal performance all three factors must be scaled up in tandem.

Unsupervised Classification Approaches

Many datasets do not include annotated targets or subsets of the data cannot be used as a prediction set. In unsupervised learning, the algorithms try to to extract patterns from this data which consists mostly of similarities or differences in the training set.

A common unsupervised approach in natural language processing for extracting keywords or similarities from text is called Latent Semantic Analysis.

Latent Semantic Analysis (LSA) [32] is an information retrieval method in text based on singular value decomposition. The algorithm first transforms every word from a sentence into a matrix M. This matrix can be dismantled into a diagonal matrix S and two orthogonal matrices U, V through

$$M = U \cdot S \cdot V^T \tag{3}$$

S contains singular values which can be easily converted to eigenvalues. These are used to process a query on the vector space to find subsets of M which fullfills the query by applying the cosine similarity. This whole method is quite expensive in computation time because of the necessary decomposition and dimension reductions [33].

There are many more unsupervised approaches for emotion detection and information retrieval in text [34], [35] but they will not be part of the following comparisons and benchmarks as they also exceed the scope of this thesis. Furthermore, no significant improvements and differences are expected in comparison to the presented unsupervised approaches and also the shown supervised classification methods due to the scaling laws for neural networks.

2.2.2. Classification in Audio

Similar to classification in text as shown in Chapter 2.2.1 there are some approaches to supervised and unsupervised methods. First of all an encoding for the audio is necessary to ensure the extraction of the necessary features.

Fourier Transformation

To break up the audio signal into multiple atomic sinusodial functions, a fourier transformation and analysis can be applied. For each frequency $\omega \in \mathbb{R}$ from the given signal, the transformation yields a coefficient d_ω and a phase φ_ω from which a sinusodial oscillation can be retrieved. Looking back at the equation from Chapter 2.1.3 a normalization with regard to their amplitude is collected by assuming $A = \sqrt{2}$.

$$g(t) := A \sin(2\pi(\omega t - \varphi)) \tag{4}$$

For each frequency and phase the normalized function $\cos_{\omega,\varphi}(t) : \mathbb{R} \to \mathbb{R}$ is given by:

$$\cos_{\omega,\varphi}(t) = \sqrt{2} \cos(2\pi(\omega t - \varphi)) \tag{5}$$

Due to the idea of approximating the original audio function with sinusodial waves, a first approach to the fourier transformation can be obtained by maximizing the similarity between both signals.

$$\begin{aligned} d_\omega &:= \max_{\varphi \in [0,1)} \left(\int_{t \in \mathbb{R}} f(t) \cos_{\omega,\varphi}(t) \, \mathrm{d}t \right) \\ \varphi_\omega &:= \operatorname{argmax}_{\varphi \in [0,1)} \left(\int_{t \in \mathbb{R}} f(t) \cos_{\omega,\varphi}(t) \, \mathrm{d}t \right) \end{aligned} \tag{6}$$

Bringing this into the context with polar coordinates and the additional idea to store both values d_w and φ_ω into a single complex value. This results in the common fourier transformation form given by:

$$\begin{aligned} \hat{f}(\omega) &= \int_{t \in \mathbb{R}} f(t) \exp(-2\pi i \omega t) \, \mathrm{d}t \\ &= \int_{t \in \mathbb{R}} f(t) \cos(-2\pi\omega t) \, \mathrm{d}t + i \int_{t \in \mathbb{R}} f(t) \sin(-2\pi\omega t) \, \mathrm{d}t \end{aligned} \tag{7}$$

These equations are however only suitable for non-discrete signal-processing and must be adapted to a discrete fourier transformation. A sampling period T is introduced to sample the signal in multiple time steps. An approximation for $\hat{f}(\omega)$ is obtained

$$\hat{f}(\omega) \approx \sum_{n \in \mathbb{Z}} T(f(nT)) \exp(-2\pi i \omega n) \tag{8}$$

with a time complexity of $O(N^2)$ where N denotes the length from the given signal [6, pp. 39-50]. Thus N can get quite large certain optimizations had to be developed for efficient usage in bigger datasets. For example the fast fourier transformation, with a complexity of $O(N \log N)$, uses a divide and conquer algorithm to split the signal, calculate the fourier transformation and reassemble it [36]. Also common is the short time fourier transformation which divides the audio into windows with a given size n and a hop size to determine the step size of the sliding window [6, pp. 51-54], [37].

Supervised Classification Approaches

Analogous to Chapter 2.2.1 the supervised classification approaches follow the same structure. Recurrent- and convolution-based networks are the most common approach [38], [39] to classify features from audio after retrieving the fourier representation for every signal and can also be applied to emotion. Continuing from the already presented fourier transformation and analysis, these supervised classification methods refer to the feature extraction from spectograms [40], [41] which is the visual representation of the given signal and a lightweight implementation for these networks.

In research, the first transformer based audio classification are presented [42]. The idea is an amalgamation of the idea from convolution architectures and the common transformer model.

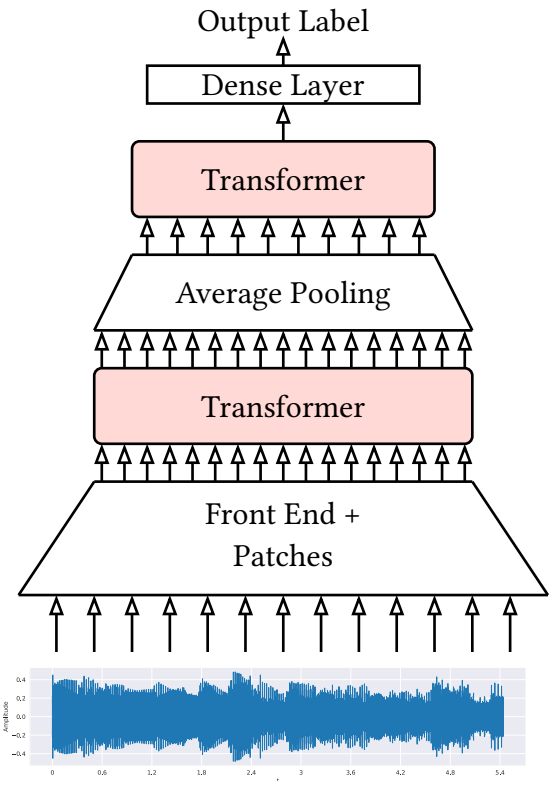


Figure 2.11.: Audio transformer model

Therefore it inputs the raw audio data without any preprocessing besides padding to the right length. As shown in Figure 2.11 the model first divides the signal into multiple fixed-length patches to feed them into a front-end layer. The patches describe very short sequences of the audio and show the resolution in which the work will subsequently be carried out. It consists of a few dense layers which process the raw audio to multiple feature vectors. These are suitable for learning in the common transformer architecture. Furthermore they are chained together with an average pooling layer, giving the convolution-like downscaling and compressing effect. This step can be repeated multiple times until the required output shape is nearly matched. At the end a decoder layer in form of some dense layers is applied to predict the resulting label. Equivalent to the already introduced convolution networks, activation functions like ReLu or softmax can be interposed between all existing layers for more stability. Nonetheless the bitter lesson from the previous chapter also applies here. For this reason and because of the already large scope of this thesis, further considerations about other supervised approaches like SVM or decision trees, are not made.

Unsupervised Classification Approaches

Unfortunately unsupervised audio information retrieval approaches are quite rare. There are a few methods which use k-means to learn from local audio features [43]. Thus they are not really suitable for emotion detection due to their low capability of detecting global features. However in spite of that there are some methods to collect local features which can assist the supervised deep learning method [44].

2.2.3. Classification in Images and Videos

Relating to the previous classification methods, the foundation for emotion detection in images and videos is quite similar. However, there exists much research to optimize the supervised and unsupervised strategies regarding the input size and computation time. In addition, video has interesting advantages and disadvantages as compared to images, which is necessary for the following methods and optimizations. The disadvantages of video mostly arise from the additional dimension time, which is introduced through videos as a time series while they can be dissected into single images.

Supervised Classification Approaches

As always, convolution models are the most common architectures for deep learning on images and videos [45]. While images can be flattened to a one-dimensional vector to process it with simple convolution filters, layers or even with recurrent and simple linear models, video processing and learning arrives with a bigger challenge. Fortunately kernels for convolution are able to have n-dimensional shapes though this leads to more expensive computation and memory consumption. The same issue occurs naturally with video transformers [46]. These follow logically the already presented architectures in text and audio classification as they transfer the researched benefits from transformer models to the vision domain. Nevertheless they scale quadratically with respect to the input length. Thus further optimizations on the input, which will be presented in the following sections, are helpful and even necessary.

Slicing offers different options to reduce the input size from the video with a variety of selection methods [47].

- **Time Exclusive Slicing:** As the most simple approach, time exclusive slicing picks one or more single frames out of the given video, resulting into a more compact version like a flip book. This selection can be done randomly or with a fixed selection of the frame indices [48].
- **Spatial Exclusive Slicing:** This method divides each frame into a grid of spatial patches. Then it chooses one of the grids (mostly the center) and picks the same patch over the full time dimension.
- **Cubicles:** Cubicles or tubelets [49] are extracted by choosing spatial patches in combination with its temporal neighbours. This is similar to spatial exclusive slicing but with more variety in the chosen grid positions.
- **Shards:** This method [50] is a combination of spatial exclusive slicing, cubicles and time exclusive slicing in which spatial patches are chosen from random or fixed and indexed frames.

Overall these options reduce the input size of the input video to a fraction of the original size on the basis of the assumption that there are unnecessary parts in each frame like the corners and that videos with more frames per second include more redundancy, such as very small changes and movements between each time step. Furthermore the basic spatial exclusive slicing is not well suited due to its inability of removing the

described redundancy. In general spatial patching can be challenging, if the person in the video is moving or the main actors are in different positions in each data sample.

Further methods for input optimizations regarding the idea of tracking movements are presented like utilizing temporal difference images [47]. It follows that deep learning architectures only learn features from the moving parts of a face and that non-moving parts might be unnecessary. According to Ekman and the presented action units from Chapter 2.1.4 this hypothesis has a good foundation. With time exclusive slicing they picked frames both randomly and with a fixed distance between the time steps and subtracted the frames from each other to create a temporal difference image for each video. Each of them shows the moving action units which were present in the data sample before. Thus it shows the action units which are normally learned by the first layers from an architecture.

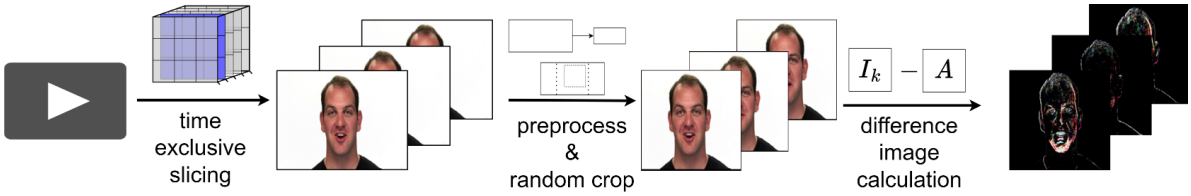


Figure 2.12.: An overview of the preprocessing pipeline [47].

This optimization method is shown by Figure 2.12 where the resulting temporal image shows only the moving parts from the video. Thus it may save some computation time but mostly it is able to save memory cost due to the reduction from 30–120 frames per input to only one temporal frame.

A similar, yet different approach is called optical flow [51], where for each video the moving parts and how they change is calculated and represented as multiple feature vectors. Here a video is represented mathematically by $I(x, y, t)$ where t denotes the time dimension as the single frames. The coordinates x and y belong to points defined in advance at which the movement is then tracked. These can be for instance the corners of the mouth, parts of the eyebrows or the eyelids. More points can be used to get a more precise resolution but also a larger input sequence. An appropriate amount would be between 20 and 30 to capture all 26 basic action units and some more details. The resulting optical flow while tracking is denoted by

$$\frac{\mathrm{d}t}{\mathrm{d}x} = I_x u + I_y v + I_t = 0 \tag{9}$$

where (u, v) represent the direction where the pixel at (x, y) is moving to. The resulting flow matrices can be used for a convolution deep learning architecture to determine the corresponding emotion. All in all there are exists many approaches to reduce the problem with high usage of memory and computation time. If these aspects can be ignored a simple 3D convolution network will cover the time dimension and can be adapted to the structure of a video [52].

Unsupervised Classification Approaches

The challenge for unsupervised classification is the feature extraction from the video which needs some kind of tracking like optical flow. Thus it has to determine which action unit is activated leading to a combination of information which then are mapped to an emotion, as presented in Chapter 2.1.4. The already shown preprocessing possibilities from supervised approaches showcase a first look about possible feature extraction, which can also be helpful for unlabeled datasets.

Appearance-based Approaches follow the idea that different facial regions are defined by unique colours and textures. In general it is approached by mapping a coordinate grid absolute to a frontal picture of a head [53] or relative to fixed marks [54]. Each face region is divided into non-overlapping patches and blocks. Each block is represented as a single feature vector and contains one facial landmark which can be clustered to an action unit afterwards. However, it observes the moving parts by colour, texture and the position in the frontal face it is sensitive to changes of light and even smaller changes of the head position. Therefore it is a suitable method for testing and methodological improvements but is not quite suitable for real life application due to these sensitivities.

Geometry-based Approaches are not well researched due to the difficulty to extracting and obtaining facial point locations [55]. However, somewhat similar to the appearance-based approaches, the idea for geometric feature extraction involves looking for changes in the static face but moreso for deplacement of the same point location due to muscular changes. In combination with the aforementioned approach, it can result in improved performance independent of lighting and outer conditions.

Motion-based Approaches are based on dense motion rather than the discrete motion from facial landmark points as presented in geometry-based approaches. Image substraction, as one subset, which was already shown during input minimisation for supervised approaches can also be used together with a combination of basic clustering algorithms like k-means [56]. Also the presented optical flow method is capeable of being used for unsupervised learning adapting it to quadtree decomposition for matching the important moving parts.

In general these feature extraction approaches use k-means with an alignment to time series. Often used is dynamic time warping, which is a distance metric and useful for time series with different lengths. Thus it has a drawback due to not always satisfying the triangle inequality which is usually an important property of these metrics. Further optimizations lead to a method called dynamic time alignnment kernels [57] defined by

$$\tau = \max_{Q} \sum_{c=1}^{l} \frac{1}{n_x + n_y} (q_{1c} - q_{1c-1} + q_{2c} - q_{2c-1}) \kappa_{q_{1c} q_{2c}} \tag{10}$$

where $\kappa_{q_{1c} q_{2c}}$ represents the kernel similarity between single frames from two video inputs. A possible kernel could be $\kappa_{q_{1c} q_{2c}}(x_i, y_i) = \exp\left(\frac{1}{2\sigma^2} \|x_i - y_i\|_2^2\right)$ with σ^2 as the average distance to the 10% of its nearest neighbours [58]. Clustering the features is not limited to k-means but is the most common approach. Generally, the research in time series clustering has developed

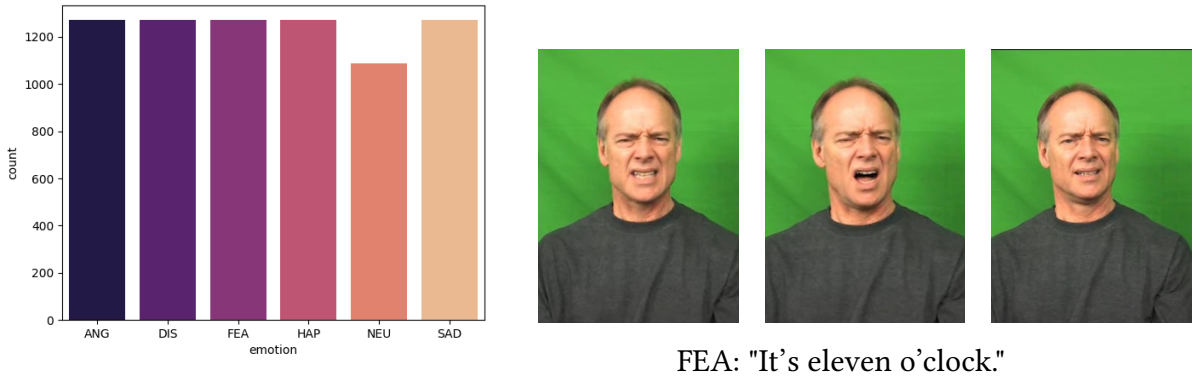
over the last years bringing different metrics and algorithms to the front and can be adapted with little to no effort [59].

2.3. Datasets

For the multimodal analysis and the supervised approaches in learning, a dataset consisting of at minimum video with audio is needed. From this state the text transcription can be retrieved during preprocessing. During this thesis, two different datasets are used which fulfill these requirements. The following datasets are based on the Ekman emotion model with some changes.

CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset) [8] consists of 7,442 clips with 91 different actors featuring their facial and vocal expressions. It covers every of Ekmans emotions: *Anger (ANG), Disgust (DIS), Fear (FEA), Happy (HAP), Neutral (NEU), Sad (SAD)* with the exception of *surprise (SUP)*, because it can relate to any other emotion with a rapid onset. additionally there are a few annotations to the intensity of the sentiment, but it is only a small percentage regarding the amount of data. Therefore it will not be used for further evaluation in this thesis.

The actors recorded twelve different sentences for each type of emotion such as “Don’t forget a jacket.” or “I wonder what this is about.” which have a semantically neutral connotation. These snippets are between 50 to 80 frames long with a framerate of about 30 frames per second.

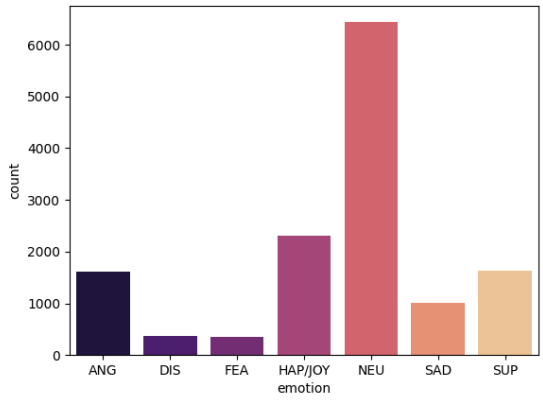


FEA: "It’s eleven o’clock."

Figure 2.13.: Distribution and example of the emotion presented by the CREMA dataset

As shown in Figure 2.13 the distribution of labels is uniformly distributed with the exception of neutral. This won’t leed to any problem, because the actual sentiments shown in the visuell data are very strongly acted so there are clear features available.

MELD (Multimodal EmotionLines Dataset) [60] is an annotated collection of short sequences from the series “Friends”. In difference to the CREMA dataset or most other multimodal datasets for sentiment analysis like IEMOCAP [61] the given data has many multi-party conversations where there is more then one participant active and present in most parts. This leads to a more complex and challenging feature extraction but approximates more closely to reality.





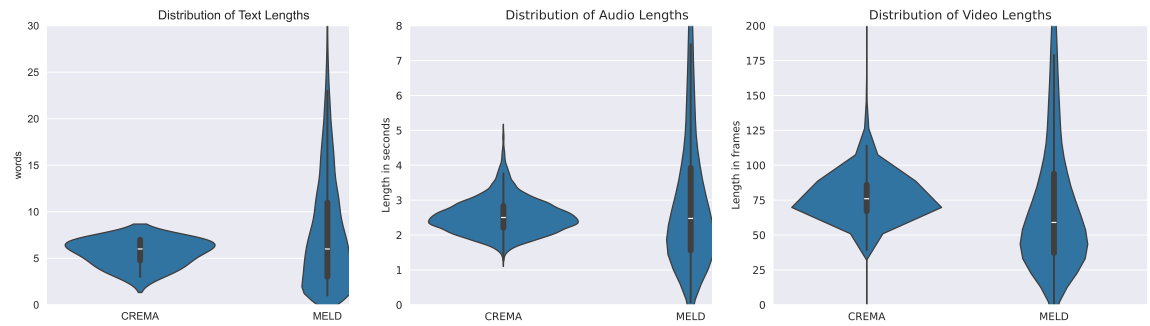
JOY: "You are a genius." (Chandler)

Figure 2.14.: Distribution and example of the emotion presented by the MELD dataset

Figure 2.14 shows the uneven distribution of the different sentiments, which can lead to an imbalanced training. Nevertheless it is a quite large dataset with 13,708 annotations which includes in addition to the emotion, the character as the speaking person which can be used to analyze the different features depending on gender or age.

Data Statistics and Analysis

Looking at the data distribution presented by the violin plots in Figure 2.15, a clear pattern emerges when focussing on the length distribution of text, audio and video. Wider areas along the x-axis are generally showing a higher occurence of the length. Further the range is shown by the y-axis. The CREMA-D dataset includes many shorter sequences with little to no scattering leading to an average of about 2.5 seconds of content. In contrast the MELD dataset includes a variety of different length of up to or more than five seconds of material thus showing a mostly similar average of time. In conclusion CREMA propagates the lack of variety in sentences as shown in (a) through the video (c) and audio (b) content due to their limitation of only twelve different sentences. With the greater real-world applicability of MELD, it features a high variety of different content lengths, which could lead to an increase in the need for more data preprocessing like padding and a larger model architecture depending on the input size.



(a) Text (b) Audio (c) Video

Figure 2.15.: Distribution of the different modal lengths for each dataset.

Challenges

As oreviously described before, the datasets bring about different challenges and requirements for the model architectures. One such challenge or obstacle, which will be important for the later evaluation, stems from the CREMA-D dataset. It consists of the twelve sentence which are acted in each emotion. Therefore it is not possible to predict any reliable label for the emotion detection in text. A general challenge for the MELD data is the imbalance between each label. As the neutral emotion is by far the most common annoation the models will mostly predict only this category, achieving a high accuracy regardless. Furthermore, due to the imbalance in the actual data sizes and as before explained with Figure 2.15, the MELD dataset covers a high range of different input lengths for each modal. Due to this reason, some adaptation to the maximum input length for both datasets is needed, or alternatively, some kind of input compression such that in the end all data entries have the same length for training. Another big challenge which will be discussed further in Chapter 4 is the non-scientific recording of the videos from MELD. As they stem from a television series the main acting person, where the model has to detect and retrieve the corresponding emotion, is not always at the same position in the video snippets. Additionally, background characters visible in the data samples could be interpreted as important by the learning model. Generally speaking, there are many unnecessary details in the MELD videos which are obstructive for correct learning. The last and biggest challenge can, yet again, be found only within the MELD dataset. It is highly imbalanced with over 50% of the labels being neutral and only small occurences of disgust and fear. Therefore a good training would be complicated as the model will mostly predict neutral and reaches a high accuracy without looking at the features.

Chapter 3.

Related Work

Emotion detection represents a field of research which has developed much over the time. An increasing number of multimodal emotion detection approaches and their resulting evaluation have emerged in the last few years. In the following chapter, multiple explorations are presented with their idea and the result. Also a comparison of the different approaches will take place while presenting the ideas shown in the works.

3.1. Audio and Video Based

Temporal Multimodal Fusion for Video Emotion Classification in the Wild by Vielzeuf et al. [62] builds a framework on audio and visual features where it tries different combinations and different scorings of the features. They use convolution models as a deep learning architecture and the OpenSmile toolkit as an audio feature extractor to retrieve the six emotions from Ekman and, additionally the neutral state in a supervised manner. The visual information is extracted frame by frame and as a consecutive video within two different models. For training they used the Acted Facial Emotion in the Wild dataset [63] which consists of different movie scenes and is comparable with the presented MELD dataset. In total Vielzeuf et al. manage to score an accuracy of 58.8% with their combination of weighted means but with no information about how they set the weights.

Multimodal mixed Emotion Detection [64] presents a similar method and implementation to the work presented before. Thus, they create features from hand gestures and body movement aswell which are then forwarded into a support vector machine. To extract information about the visual and audio data, they used premade frameworks like OpenEar similar to OpenSmile and FaceAPI which allows better tracking for faces and head rotations. additionally they used an infrared sensor from Microsoft for the gestures and moving parts of the body. Furthermore, they classified the six Ekman emotions like the previous paper but without any neutral emotion. As results they presented an accuracy of 77.5% while using data from the hand gestures, 65.2% for the body modality and 94.3% for using information from the head region. In total by combining all features they manage to obrain an accuracy of 96.6% which is quite promising. However, they are using the IEMOCAP data which is closer to the CREMA-D set and incorporates more acted emotions into training than a more natural dataset would do which was used in the work before. Also they did not include any feature combination with weights or other methods. The high difference in the final accuracy shows the difficulty with natural moving parts or moving camera perspectives as those shown in movie scenes.

3.2. Audio, Video and Text Based

Multi-level Feature Optimization and Multimodal Contextual Fusion for Sentiment Analysis and Emotion Classification [65] uses text, audio and video as input, extracting the features with evolutionary computing based models like GA or GGA [66]. They propose a multi layer feature extraction which propagates the raw input through different layers. The first subsets of information and patterns are extracted by unimodel GAs and GGAs, and then forwared into a recurrent LSTM network to retrieve more contextual input. This procedure will be repeated multiple times and, between each step, a fully concatenation of all features is applied. Following that, the unimodal architecture gets split into a trimodal architecture. For training and testing they used the IEMOCAP database as a static dataset and the CMU-MOSI data. This set consists of opinionated video essays from YouTube split up into multiple emotionally affected segments making it comparable with the MELD or the WILD database. As shown in Table 3.1 their results show a reasonably reliable accuracy improvement when using a combination of multimodal features. In single feature usage they got a score of 69.49% for video, 77.28% in text and 68.28% while using the IEMOCAP dataset and similar results for the CMU-MOSI model.

Modality	Accuracy CMU-MOSI	Accuracy IEMOCAP
T+A	77.06%	78.85%
T+V	76.65%	79.12%
A+V	70.00%	65.44
T+A+V	80.38%	79.52%

Table 3.1.: Comparison of results. A = Audio, V = Video, T = Text.

To combine the features, they used contextual fusion in which the model trains the optimal weights for the composition. Furthermore it shows that between text and audio, no further improvement is to be expected. Thus it presents a reliable architecture for both datasets.

Multimodal Sentiment Analysis using Hierarchical Fusion with Context Modeling [67] follows a slightly similar approach to the work before. Instead of GGAs for feature extraction they used basic convolution networks trained with the IEMOCAP and CMU-MOSI databases. OpenSmile was used as a pretrained processor for audio. They present a mostly identical feature combination strategy with a fully connected layer and gated recurrent units which are used for contextual learning and to combine the features while learning the optimal weights. Overall they achieved a little less accuracy than the feature extraction with generative models but for text, video and audio in combination they recieved a score of 80.0%. Audio and video achieved 58.8%. Bimodal combinations with text got an accuracy of roughly 79% as well as the insight that there is not much room for improvement in comparison to the trimodal result. For the single model evaluation, text performed the best with an accuracy of 76.5%. The other two models bring about 50% to the evaluation which is quite low in comparison to the already presented methods.

Seq2Seq2Sentiment: Multimodal Sequence to Sequence Models for Sentiment Analysis researched by Pham et al. [68] presents an architecture based on a special class from recurrent networks. The basic idea is to first train a sequence-to-sequence model for two modalities and afterwards a combination of the already trained model and the third modal to the final result. Therefore they use the CMU-MOSI dataset and a feature extraction build on different pre-trained encoders. For text input a GloVE model [69] — which represents its own idea of encoding besides skip-gram or bag-of-word models — was used in lieu of an already pre-trained one. COVAREP [70] was used to extract multiple important features and feature representation from the audio signals and for videos a motion-based representation extraction method is proposed. Altogether they did not measure the accuracy for single modal training due to the structure of the architecture. Thus they got a score of 75% for all three modalities combined. Interesting to note are the accuracies for the bimodal detections: Text in combination with video got 78%. The other two modal permutations lay at 44% for text with audio and 55% for audio and video. According to Pham et al., the reason for these rather subpar results stem from poor video and audio feature training of the model architecture. Instead, they recommend other encoding mechanisms and a slightly different model architecture based on sequence-to-sequence for future work.

Tensor Fusion Network for Multimodal Sentiment Analysis [71] presents the method of tensor fusion for the necessary feature combination in context of the multimodality. It describes an untrainable algorithm which attemps to combine multiple tensors by a reduction to a single tensor. This results in an interplay between a recurrent and LSTM based model for each modality and the proposed tensor fusion layer. Only for text input did they show their own method for necessary feature extraction by combining LSTM cells and linear layers. Like the other works before they set their foundation for audio extraction on pretrained models like COVAREP but they combined multiple processors for different features as peak slope parameters or other high level audio information. For video they also refer to FACET as the facial expression analysis framework able to extract up to 20 action units. Now, the tensor fusion layer represents some fully connected deep layers which will learn from the combination tensor to predict the emotion. Overall they reached an accuracy of 77.1% with their unimodal network. The baseline accuracy, representing every single model, for text processing hovers around 74.8% while the visual and acoustic features score lower with roughly 66%. It outperforms the basic convolution network proposed by the first work and reaches good performance in relation to all other presented works thus being a more rare approach while using an untrainable feature combination.

A Social Emotion Classification Approach using Multi-Model Fusion [72] implements a hybrid form of convolution and recurrent networks to build a context based extraction method for all three modals. For video they combine LSTM and 3D convolution to a proper time understanding model and a simpler CNN-RNN architecture and a bag-of-words for text classification. Again for the feature extraction from the audio signals they proposed OpenSmile as a reliable framework. For the subsequent feature combination they introduce multi-kernel learning [73] which retrieves the result from each kernel generated by each modal within the CNN architecture. For comparable results they used the IEMOCAP database as well as the MOUD dataset. It is, like CMU-MOSI, a collection of YouTube videos and snippets and therefore a more real-world illustration, however, it contains over 300000 data snippets

making it rather large in comparison to all previously presented datasets. Therefore they get pretty reliable results and a trimodal accuracy of 95.88% with the MOUD dataset and 78.75% for IEMOCAP showing that a bigger dataset can lead to more exact results.

All in all the current works in the field of multimodal emotion detection do not differ all that much from each other. As shown in the following Table 3.2, the main dissimilarities are lying in the basic training architecture to train the feature combination. Thus, there are mostly some kind of reccurent architectures deposited to keep track of the context. Furthermore, most of the current works achieve indistinguishable accuracies, yet, show that a bimodal combination of text and audio brings nearly good result in comparison to the unimodal architectures. The usage of pretrained processing layers and feature extraction like FACET, GloVE and OpenSmile highlights a huge reduction in computation and necessary preprocessing time. It should be noted that no research about uni- and multimodal emotion detection featuring leading transformer arhcitectures could be found, as of now.

Author	Year	Dataset	Architecture	Fusion Method	Accuracy Bimodal	Accuracy Trimodal
Vielzeuf et al. [62]	2017	WILD	CNN	Weighted Mean	58.8% (A + V)	-
Majumder et al. [64]	2018	IEMOCAP	SVM	Linear Layer	96.6% (A + V)	-
Huddar et al. [65]	2020	IEMOCAP, CMU-MOSI	GGA, GA, Recurrent LSTM	Contextual Fusion	77.06% (T + A) 76.65% (T + V) 70.00% (A + V)	80.83%
Majumder et al. [67]	2018	IEMOCAP, CMU-MOSI	CNN	Recurrent Networks	77.30% (T + A) 77.80% (T + V) 58.80% (A + V)	80.00%
Pham et al. [68]	2018	CMU-MOSI	Seq2Seq	Hierarchical Fusion	44.00% (T + A) 78.00% (T + V) 55.00% (A + V)	75.00%
Zadeh et al. [71]	2017	CMU-MOSI	LSTM	Tensor Fusion	-	77.10%
Xu et al. [72]	2020	IEMOCAP, MOUD	CNN-RNN	Multi-kernel learning	85.36% (T + A) 92.37% (T + V) 91.14% (A + V)	95.5%

Table 3.2.: Overview of the compared works showing the bimodal and trimodal results. A = Audio, V = Video, T = Text, - = not provided / not evaluated.

Chapter 4.

Design and Implementation

Given the six emotion from Ekman explained in Chapter 2.1.1 and a neutral label, the subsequent chapter focuses on the implementation of the different supervised model architectures for each modal. This will be done for the presented datasets MELD and CREMA-D which are both challenging various aspects of emotion detection. Furthermore the important preprocessing and feature extraction for each model are described. Both the bimodal and unimodal combination are implemented showing different types of feature combination or voting and scoring methods. Figure 4.1 shows an overview of the presented implementation goal. In the following each component of this architecture will be described and discussed.

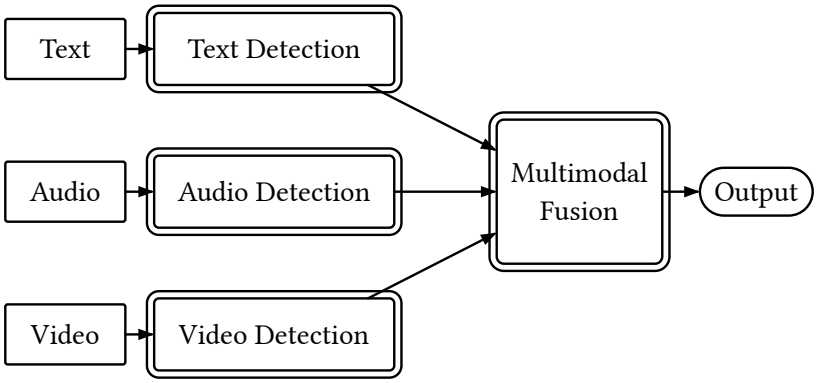


Figure 4.1.: Pipeline for multimodal emotion detection.

4.1. Unimodal Text Classification

As described in Chapter 2.2.1 transformer models are quite suitable for text classification. However, they need some sort of word embedding for preprocessing. Fortunately already existing Large Language Models are capeable of processing text as normal conversation input for giving mostly correct answer to the prompted sentences. The most common models are also pretrained on very large datasets, for example papers from scientific websites or collections of sources and articles like Wikipedia. Although needing some adapatations and further finetuning to the used datasets in this work, the following three state-of-the-art pretrained networks are used for measurement and classification.

Llama3 [74] developed by Meta presents a generative deep-learning model based transformer. It consists of over 128 thousand known vocabularies from open sources in the internet.

With more than four billion parameters and its well preprocessed dataset, they show reliable results in text conversations. As a natural language understanding model similar to ChatGPT or other various models, Llama provides contextual understanding in a chat like conversation with the user. In contrast to most big architectures, Meta decided to make their model open-source with all the pretrained weights and different interfaces to run the product locally.

Multi-EmoBERT [75] was originally applied to detect fake news with the support of emotion classification. Thus it is more an architecture than a pretrained model, offering the possibility to train and finetune the architecture with custom text data. As already briefly explained in the background of text preprocessing, BERT is a bidirectional encoder with an underlying transformer developed by Google in 2018. The multimotion aspect gives multiple labels and their probabilities for a given input leading to a higher variety and more options to choose the correct label.

T5 Model [76] is a text based transformer model which settles on text-to-text transfer learning, following the idea from Llama 3 and, likewise, also trained on open source crawled data from the internet. The advantage to the already presented architectures and implementations is an already existing open source finetuning on sentiment and emotion detection reducing the effort for necessary preprocessing.

Architecture Overview

As previously explained, an emotion classification with text will only be done with already pretrained models therefore they will be considered as a black-box but with some options of finetuning.

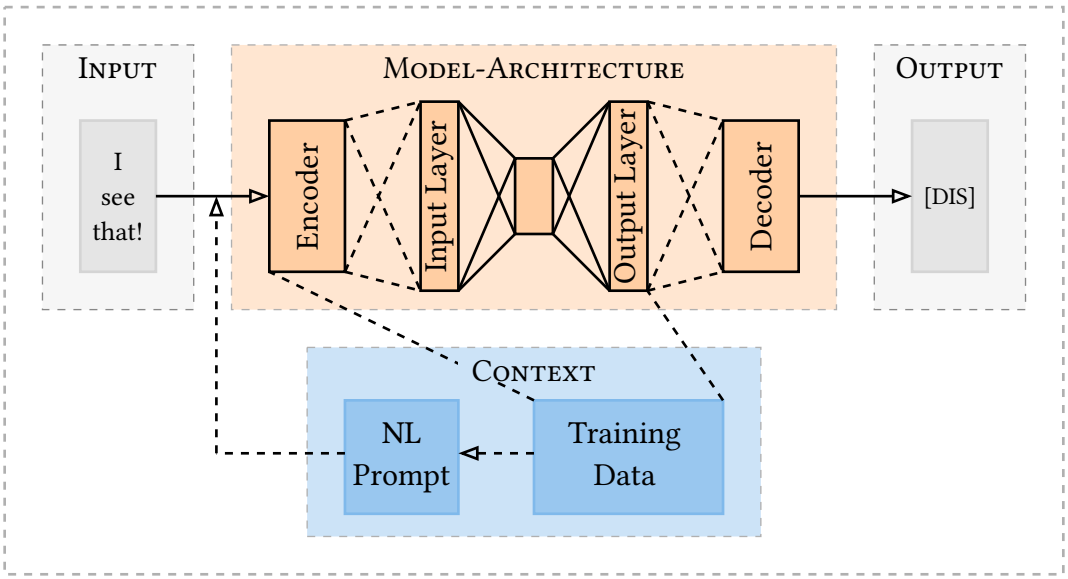


Figure 4.2.: Structural procedure for emotion classification and possible influences on pretrained text-to-text models.

Figure 4.2 shows the sequence of the processing which can be adapted to the three shown transformer models. In general such models require an encoder with a tokenizer and decoder to understand the given natural language input further to output human understandable la-

bels for the corresponding emotion. In between, their model architecture is mostly defined by their adaptation of a transformer and is already well trained which results in a set of pre-defined weights. For finetuning these models offer two fundamentally different methods: prompt engineering and post-training tasks, which allows adding context to the architectures or weights. Some text-to-text models try to simulate human-like chat conversations with contextual memory over a talk. Foremost it gives the opportunity to define tasks which will be fulfilled by the artificial opposite. It introduces a scope to feed in the available training data to the model by embedding it into the chat prompts and define the task to learn for a limited period of time. This leads to the problem that the given knowledge only exists within the scope of the current chat window and is permanently deleted afterwards — the feeding must, henceforth, be repeated every time. Therefore the second method for finetuning shows a procedure to train on top of the current pretrained weights and the encode layer with a model specific interface. It offers a more reliable context injection in comparison to the prompt engineering method and provides transparency about the learned content.

4.1.1. Preprocessing and Prompt Engineering for LLM

Model	Response	Type of Response
Llama3	I'm glad I could share the excitement with you! It's always thrilling to spot a clever joke or play on words. If you're ready for more, I've got a few more where that came from!	Natural Language Conversation
Multi-EmoBERT	anger: 0.0098 disgust: 0.0050 fear: 0.0113 joy: 0.3621 sadness: 0.0121 suprise: 0.0564 anticipation 0.5360 optimism 0.4288 pessimism 0.0121 trust 0.0330	Probabilities for every emotion
T5 Model	joy	emotion label

Table 4.1.: Answers to the prompt "I see that!" from the three text-to-text models without finetuning and any context.

Fortunately all of the three given models implement their own language understanding system. Therefore it is not necessary to postpone and train a personal word embedding. Mostly they rely on some variants of bag-of-words to encode the vocabulary tokens. In all three cases the models consist of an encoder and a decoder layer for the text-to-text understanding and

their specific transformer based architectures to give them their expected user-oriented functionalities. Thus, just inputting a sentence to retrieve the emotion label will not work in most cases due to the attempt to imitate real-world conversations. Looking at the results for the input sentence “I see that!” the models show the listed responses in Table 4.1.

Llama 3

Llama’s response shows the conversation-like answer leading to the need of some finetuning. Fortunately the model has contextual memory which can be used to predefine a task with natural language for the architecture. In this case the assignment for better understanding is given and repeated with an example after every prompt to refresh the contextual memory. Further, as shown in Figure 4.2 and explained before, the training data can be embedded in the progression of the conversation to give the model a better overview about the data which can lead to a higher accuracy.

With this task the model outputs a more sensible answer in the right format for the same example sentence. This prompt can be changed slightly to feed in the contextual information about the training data without much effort as shown in the following Listing 4.1.

```
# Define initial task
message = [
     {
          'role': 'user'
          'message': "Giving the following seven emotion..."
     }
]

# Context generation
for sentence, emotion in train_set:
     message.append(
          {
              'role': 'user',
              'content': sentence + " " + emotion + "\n"
          })

# test the data with a copy of the context
for sentence, emotion in test_set:
      prompt = message.copy().append(
          {
            'role': 'user'
            'content': "The prompt is: " + sentence
          })

     response = ollama.chat(model='llama3', messages=prompt) # results
```

Listing 4.1.: Short overview of the prompt input to the Llama model with its contextual feed.

A possible prompt which is used later for the evaluation is:

> "Giving the following seven emotion with their abbrevation: Anger (ANG), Disgust (DIS), Fear (FEA), Surprise (SUP), Sadness (SAD), Happiness (HAP), Neutral (NEU). I'll prompt you a sentence and you will answer with the corresponding emotion abbrevation. For example I'll ask: "My day is awesome" and you're answer would be "[HAP]". Don't forget the brackets. Sentences, or questions like "okay" or "lets go" are also sentences where you have to guess the emotion. Alright?"

Multi-EmoBert

However as mentioned, it is a bit more complicated with the Multi-EmoBert due to the lack of any context based chat functionality as they already assign emotion labels as a response. Looking back at Figure 4.2 and the notes before, there are possibilities to train the architectures — in particular the encoder and the weights — on top of their pretrained values with a custom set of data. This opens up the opportunity to finetune the models and enhance the results. The data from the training dataset has to be processed to the specific format which is already offered by an interface from the source model. An application of a threshold to validate the correct emotion label is necessary as it outputs all probabilities for every sentiment showing another variable factor as shown in Listing 4.2 which has impact on the accuracy. A lower value might increase the chance to include the correct emotion in the set of all sentiments which are guessed with a higher possibility. Thus the amount of false positive emotion detections must be considered as well. With a higher threshold the possibility to guess the correct label might be lower but also the magnitude for all misclassified labels is lower. If there are no classifications which reached the minimum possibility, the sentence is tagged as neutral. All sentiments given by the model which are not part of Ekmans six basic emotions or neutral are ignored, as they are based on the original training data and have much lower possibilities after retraining the model with the custom dataset excluding these additional sentiments.

```
1  def choose_emotion(P, th):
2      L = [ ] # list for output
3
4      # loop through every possibility
5      for e in P:
6            if P[e] >= th: # if matches the threshold
7                 L.append(e)
8
9      # if none of the emotions reaches the threshold
10     if  L.is_empty():
11           return ["neutral"]
12     return L
```

Listing 4.2.: Simple algorithm for choosing all emotions from the possibility-space P with an threshold th.

T5

The T5 accepts both methods for finetuning thus prompt engineering does not seems to be mandatory due to the already labeled output. However the combination might show interesting results in comparison to the other two architectures with only one of the two finetuning methods. In comparison to the Multi-EmoBERT model the T5 will not offer a direct interface for dataset conversion and an active method for training but it has an low-level connection through the PyTorch library from python, which allows training like a regular self-implemented model but with respect to the pretrained weights. In addition the documentation gives some functions to enable regular training. As a PyTorch submodule it includes a loss function based on the cross entropy loss function to compare the resulting token values. Also the encoder and tokenizer part is able to recognize new tokens and adapt itself without any further external influences or preprocessing.

4.2. Unimodal Audio Classification

In contrast to text classification, not many pretained and large networks available. Many different architectures and model types are exist as described and shown in Chapter 2.2.2. Fortunately multiple feature extraction methods and tools can be used, providing a large range of different information on the basis of the fourier transformation. Furthermore, a custom encoder and decoder mechanism must be implemented for audio understanding and reliable labeling.

Preprocessing and Feature Extraction

OpenSmile extracts a variety of valuable information from an audio segment giving a good overview about necessary components which are useful for the corresponding set of emotions. With the knowledge from Table 2.1 a small cutout of all extracted features is presented in the following sections. This excerpt shows the significant types of information which are mostly supported by similar features.

Loudness describes the volume of an audio and is able to describe firmly two different sets of emotion, calmer sentiments like sadness or neutral and more powerful ones like anger or happiness. In general louder voices lead to the assignment of an aggressive emotion while calmer audios represent harmonious or more melancholic sentiments of a person.

Alpha Ratio shows the relation between frequency above and below 1000 Hz. In support of loudness, this ratio analyzes vocal volumes and not only the volume of the audio sample itself. Additionally, some information about timbre can be obtained from the data.

Hammarberg Index [77] can be briefly described as the intensity difference between the maximum intensity of the signal in a lower frequency set and the maximum intensity in higher frequency thus giving an energy distribution measurement for the given audio. Similar to loudness the energy index normally can be mapped to the two sets of emotion.

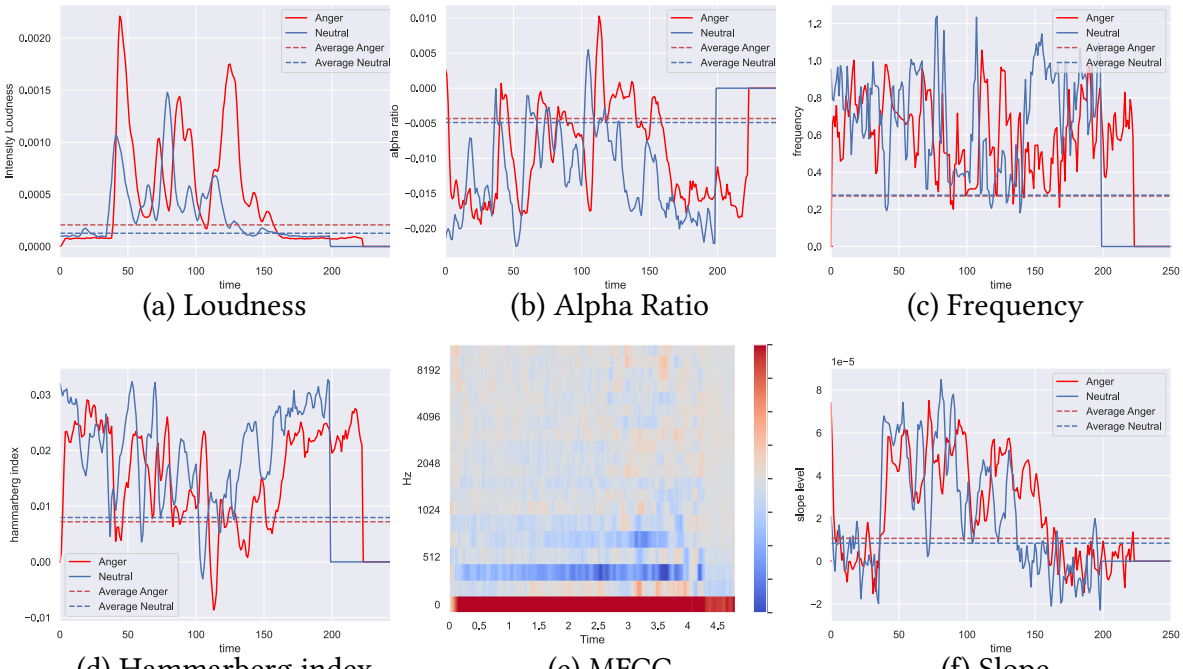
Slope measures how fast a given audio signal tries to reach high frequencies leading to different possible assumptions about pitch changes and pitch ranges which are important for several types of emotion like fear or disgust, as shown earlier in Table 2.1.

MFCC (Mel Frequency Cepstral Coefficients) are a compact representation of frequencies where different pitches are visible. They are often used for speech synthesis and other detection methods due to their high information value. In general, features like the speech rate which is important for emotion as for example fear or sadness can be obtained.

Jitter is a loss measurement while working with digital audio signals and shows if there is noise in the data while comparing frequencies and oscillations. However, it is in this case not necessary for emotion detection, nevertheless, it is generally helpful for noise detection in signal processing.

Frequencies are a low-level information source for the feature extraction. While MFCC gives a compact overview, these extracted frequencies show raw information about the audio. Thus, as already described earlier they are important for all kinds of audio classification and their information retrieval.

More features are also present resulting in a matrix with 20 different types of information which can be processed by the various network architectures. Further these layers include redundancy in multiple formats and representation for better learning and feature extraction within the models. A comparison of the subset of traits from the audio shows the changes in their data. Figure 4.3 visualizes the presented excerpt of the features which have been extracted by OpenSmile.



(a) Loudness (b) Alpha Ratio (c) Frequency

(d) Hammarberg index (e) MFCC (f) Slope

Figure 4.3.: Cutout from the different OpenSmile features comparing all anger and neutral audio samples from CREMA-D.

Each plot shows the changes of the metrics as a time series for the more powerful emotion anger and the neutral baseline. To retrieve this overview all audio data for each of the two emotions from the CREMA-D dataset are added together and the average was drawn to retrieve an average signal and to compute these features with OpenSmile. Visible in the loudness-plot the red lines show the visualization of anger. It is in general higher than the neutral plot which is shown as the blue color. Furthermore the subfigure shows that furious emotions start much louder in the main than the baseline data. This is underlined by looking at the average drawn out of the signals. Here also the main loudness from anger is much higher in contrast to neutral emotions. As shown later, it is also the highest difference between the averages leading to the assumption that loudness is a quite important factor and feature for the differentiation between these two emotions. Similarly, a look at the alpha ratio (b) shows a higher average from anger in comparison to neutral. As already explained, this ratio also shows some information about the voice volume and the timbre which further supports the theory about the importance of loudness for this comparison. Interestingly, the high amount of negative values of the alpha ratio show that there is more energy in the lower frequencies than in the higher frequencies. In this case lower frequencies lie between 50 Hz and 1000 Hz. To compare these values, a male voice generally has a mean of 111.8 Hz, while the lower frequency mean of a female voice is 161.3 Hz [78]. Its showing the reason for this distribution and values in the ratio. This is also supported by the distribution of frequencies (c). Even though it looks unequally distributed, the average for both anger and neutral is nearly the same. The very small difference might show a contrast between the sentiments like loudness or other high-level information due to its low-level representation where most features are obtained from. It is possible that small numerical issues, such as instability in calculation of the averages or the difference can be canceled out with much more data. For this overview the CREMA-D dataset just offered round about 1000 samples for neutral and slightly more than 1200 samples for anger out of the training set.

Expecting from the Hammarberg index, anger should show more energy than the neutral as already described in the alpha ratio. However, as visualized in (d) from the overview figure the neutral average is visibly higher than the calculated value from the other sentiment. This may stem from different reasons related to the recorded voice quality and timbre — sometimes from more detailed audios with less noise. A softer timbre might also lead to these results. Finally the vocal pitch could influence the results with thtese small yet measurable differences.

Interestingly, the MFCC plot (e) shows the difference from anger and neutral emotion by subtracting each MFCC value stream from the other. Each colored cuboid represents the intensity of the mel coefficient measured in decibels. Transparent areas show that there is no difference between both values. Reddish parts describe that the mel coefficient for the anger emotion is higher than the coefficient for neutral. On the contrary bluish colored cuboids represent higher values for neutral voices. Looking back at the resulting plot in Figure 4.3 there is a stark difference in the low frequency ranging from 0 Hz to 150 Hz. As already described this is the interval where the average pitch for male voices lies showing the more active and relevant parts of the plot. This shows that anger voices have a much higher mel coefficient than the neutral audios meaning that the recording contains much more energy like loudness in contrast to the baseline. The mel coefficient is somewhat more present within the higher frequencies in the neutral emotion, thus the activiy by the

acting voice in the signal is reduced. The values might occur due to noise or compression artifacts in the recording data. While listening to the audio samples a constant white noise can be observed which encourages this assumption. In some examples, this sound could be the same for both audio categories leading to the colorless areas in the presented plot.

Finally, the cutout the slope (f), shows similar to the Hammarberg index and alpha ratio, a small but observeable difference between anger and neutral. This information about the pitch range and change shows that there are some important aspects within emotion classification. Also it shows, that there exists trainable features for the neural network architectures. Negative values are usually an indication of pauses between words and sentences which can be important for measuring the pace of the speaking person or other information like spoken words per minutes.

Overall there are many more observeable features extracted by OpenSmile which covers many aspects of audio signals. There are also differences visible between each emotion and the baseline neutral. However, some of these are quite small but in some ways short patterns exist. Also as already shown there are some redundancies within the choice of extracted features which will not be a problem and might result in better training. All feature combination which were shown before in Table 2.1, are available and measured by OpenSmile and also shown in the given cutout from Figure 4.3. Loudness shows changes in the intensity, MFCC is important for the speech rate and articulation. With the alpha ratio the voice quality can be observed and slope shows features for the pitch average, range and other corresponding changes. Every plot and feature contains many visible zero values. Generally these exists at the end of the signal series due to padding with zeros and are the reason for the sudden constant progression of the time series. A neural network would not have any problems with these padded values and normally only tries to learn from the non-zero features.

Model Architectures

As already presented there exists a lot of different architectures which are able to learn and train from audio data.

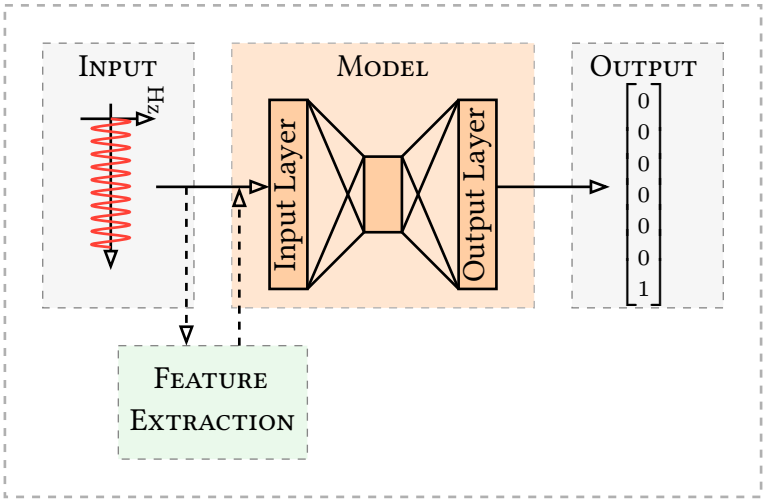


Figure 4.4.: Structural procedure for emotion classification in audio with a possible pre-trained feature extraction.

Most commonly they consist of some kind of recurrent and bidirectional neurons or in general some context understanding which is important for time series training. Similar to Figure 4.2 which shows an abstract visualization of the text to emotion processing pipeline, Figure 4.4 presents a small overview of the necessary components for the detection in audio signals. Starting with the input, it is possible to train from it without any external feature extraction. Thus only the audio transformer which was presented earlier is currently capable enough to generate reliable results. In general the convolution based architecture mostly needs a predefined feature extraction for better and faster learning results. These two models are the main models implemented in this thesis for emotion understanding in audio. Each architecture will output the calculated logits which represent the possibilities for each label to be correct. These are afterwards used to map them on to one-hot vectors showing the corresponding emotion as positive (1) and negative (0). This might also allow multi-label classification but is not introduced further in this thesis due to the limitation of the scope. For the optimization task of the models the cross-entropy loss is used for each to calculate the error between the resulting and softmaxed logits with the given one-hot vector target. Table 4.2 shows, as an important overview, the amount of trainable parameters for each model. Generally the three presented text models implement round about 4 billion parameters or more each, however they are already mostly pretrained. Both audio architectures are well comparable and within a reasonable range to these models allowing less noise due to higher model differences.

Architecture	Trainable Parameters
Convolution	2,176,474,377
Audio Transformer	1,394,574,473

Table 4.2.: Amount of trainable parameters for both implementations.

4.3. Unimodal Video Classification

Due to the feature extraction methods in audio emotion classification, which output multiple matrices with their corresponding values, the detection in videos will not differ much. Thus as already discussed the problem with the additional time dimension within the videos still exists and needs some preprocessing for improvements and optimizations to reduce the computation time while preventing any avoidable loss in training and evaluation.

Preprocessing Methods

As already discussed earlier there are several methods and ideas for preprocessing the video samples before feeding them into the networks. The simplest idea but most expensive in computation time would be to just use the full length with all available frames. The CREMA-D dataset would support this idea without much performance issues as its highest amount of frames in a video is 125 and an average of 75 frames but looking at the MELD dataset shows the significant difference with more than 200 single images in one video. A possible way which is also easy to implement would be to just extract important parts of the videos to represent

all the most significant features and movements of the speaking person. In general this is hard to track and measure without any further extraction or classification of the material. Thus an easier way is to pick every n-th frame out of the given videos instead of every frame. This will reduce the memory costs by the given picking constant and reasons by assuming that a video with about 60 frames per second must include a bunch of redundant information which are skippable. Due to the different lengths of the videos, this idea will generate different amount of important cutouts. Therefore padding with zeros would be important and in the average case there will be as many frames as the longest video divided by the constant would have, leading especially to many zero values within the MELD dataset due to the inbalanced distribution of the lengths. Adapting to this problem a further development would consist of a fixed amount of picked frames with different distances and intervals inbetween each other. In this thesis, 30 cutouts were randomly chosen from each video, which should be enough to represent the actual content and information flow without much loss. This constant nearly represents the average video length divided by 2.

Premade feature extraction methods like FACET which directly look for for information about the presenting face and acting persons in these videos are suitable for scientifically recorded datasets like CREMA-D where only one person sits directly in front of the camera. Due to the usage of the MELD dataset, which includes more real-world data in which some video cutouts show additional people or the actor is not always situated in front of the camera, FACET, or other premade feature extraction methods regarding videos, cannot be used for this kind of data.

In total there will not be much preprocessing besides randomly picking 30 frames out of each video. As usual the data will be normalized which will lose information about the color and just keeps the grayscale information in the cutouts. This will also force the model to learn features about the moving parts in the faces specifically. In this case every color is given by a separate channel for each RGB (red, green, blue) value. This should not make much difference but can give the model some colored reference points.

Another problem which occurs before the video can be forwarded into the model is the actual size of a single frame. Generally for example the vision transformer model is tested with small videos and images with a size of 16×16 pixel [79] which cannot represent the given data from both datasets. Within CREMA-D all data samples have the resolution of 480 pixel width and 360 pixel height. This results in over 100,000 values per frame, however in comparison to the MELD dataset it is still not as problematic.



Figure 4.5.: Example cutout from the MELD dataset with marked active speaker.

These videos in the set are much larger due to their origin as samples from a television show made for bigger screens. Therefore they have a resolution of 1280 pixels in width and 720 pixels in height which is nearly sevenfold the amount of values per frame and video in comparison to CREMA-D. The only solution for this problem is to crop the videos to a workable size without any loss of information. CREMA-D offers a realtively easy approach thanks to the static recording of the speaking person which is generally in the center of the video with some surroundings from the background which can be cut off. As already shortly presented in Chapter 2.3, there are several people within the MELD videos besides the speaking actor. Also as shown in Figure 4.5 the position of the observation target differs from data sample to data sample which is marked red for better visualization. There are two necessary steps available to extract the active person automatically as shown in Figure 4.6 and as an example visualized in Figure 4.7.

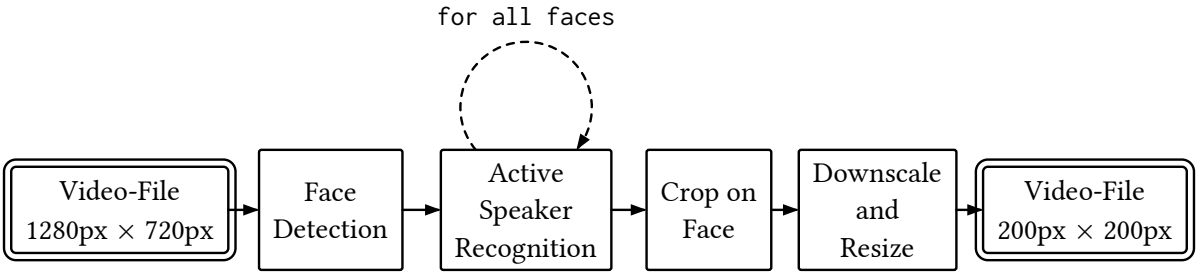
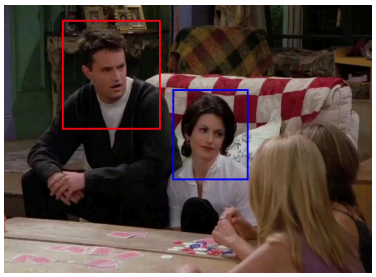


Figure 4.6.: Preprocessing pipeline for videos.

First all people within the video have to get detected by an algorithm. Fortunately, computer vision libraries like OpenCV for Python are able to recognize faces in images and therefore also in videos using pretrained machine learning algorithms. In the second step the active speaker must be found. Libraries, which are able to analyze if the given audio matches to the lip movement from a person, also already exists. Combining those methods, all unnecessary parts like background actors can be cropped out of the videos which leaves only the important active part in a smaller video size.



(a) Face recognition. Red = Active Speaker, Blue = Background Character



(b) Cropped video after active speaker recognition



(c) Resized and downscaled video

Figure 4.7.: Example for the video cropping preproceesing pipeline.

At the end a futher downscaling is applied on both datasets regarding the relative face position using, again, the face recognition capability from the computer vision libraries. This leads to a video format of 200 pixels in width and height as shown in (c) from Figure 4.7 reducing

the computation and memory cost by a lot. Visible artifacts resulting from camera shake and unclear face positioning follow as a result of this downscaling. Due to these issues the machine learning algorithms and libraries which are used are sometimes not as precise as the human understanding of face recognition. Some loss in video quality due to the downscaling which could lead to a depletion of subtle movements in the given face region. But with this preprocessing a model should be able to recognize patterns and similarities which was mostly impossible before.

Model Architectures

Similarly to the proposed audio architectures, the following structure for the classification in video will not differ that much from each other. Thus for better performance the preprocessing will be mandatory for the convolution based model and the video transformer architecture while the OpenSmile extraction was optional due to a possible understanding of the raw audio signal. The following overview in Figure 4.8 shows again, as for text and audio, an abstract architecture about the necessary steps for audio classification.

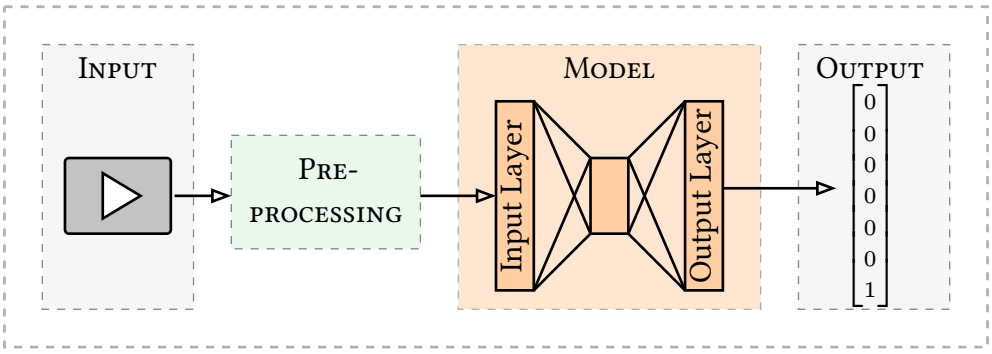


Figure 4.8.: Structural procedure for emotion classification in video.

As explained and proposed before the video will be preprocessed first where it is downscaled and cropped generally. Especially the data from the MELD set runs through the speaker recognition pipeline to filter out most of the unnecessary parts from the samples. In both datasets the amount of frames is reduced to thirty under the assumption and aim to reduce redundancy and to save computation cost. After the normalization step which is still part of the preprocessing the videos are fed into the model architectures. As the input has a size of $(\text{batch size}, 3, 30, 200, 200)$, describing the $(\text{batch size}, \text{channels}, \text{frames}, \text{width}, \text{height})$, the convolution based approach uses 3D kernels to approach the trainings loop on the given batch regarding the additional time dimension. In contrast the audio convolution model only uses 2D convolution due to the dimensionality of the structure from these time series. Every block in the network consists of a layer norm, the actual weighted convolution layer and a max-pooling. After these blocks some linear densely connected layers follow for further downscaling of the remaining shape and the final mapping to the expected class shape which is in this case and also in the case of audio classification $(\text{batch size}, 7)$ representing each emotion as a one-hot encoding. This is also visible in Figure 4.4 and Figure 4.8 while looking at the output block thus the model first outputs the logits as the likelihoods for each class. An argument of the maxima (arg max) is able to compute the corresponding label from the prob-

abilities. Furthermore the transformer model is an adaptation of a vision transformer which only accepts single images [80]. The implementation is currently available[1] as an open-source trainable pre-build architecture which only has to be integrated in the same training loop as for the convolution model.

Architecture	Trainable Parameters
Convolution	2,504,047,929
3D Vision Transformer	1,266,615,309

Table 4.3.: Amount of trainable parameters for both implementations.

For comparison, Table 4.3 shows the amount of trainable parameters for each architecture. Looking back at Table 4.2 the convolution model has a mostly similar size to the other audio based convolution model. Also in comparison to the pre-trained text models they are quite a bit smaller but they should have enough parameters for a reliable result. The 3D vision transformer architecture features a lower amount of trainable points in the given model. Thus as proposed in the original paper [27], which introduced these kinds of architectures, they generally will perform similar or sometimes even better than traditional models with less parameters.

All models will be trained in twenty epochs with a 90/10 split for training and testing. This can give an overview about the efficiency of the architectures and also about the consistency and reliability of the results. Normally these models have to be trained over months for realistic and nearly perfect results but this will exceed the scope of this thesis work. Further the cross entropy loss is calculated for the gradient descent. The Adam optimizer is utilized for further optimizations in the calcaulation and uses a learning rate of $\mu = 0.00005$.

4.4. Multimodal Classification

Here the bimodal and trimodal emotion classification are unified as they will not differ that much in their structure and the methods for combining the outputs from every model. As the unimodal detection used a straight-forward mapping from an input to the corresponding output some new ideas must be appended at the end of the given models. Some related works have already presented different feature combination methods such as kernel fusion or other ideas. In this case it is more difficult due to some limitations from the architectures themselves which are presented in this thesis. It is too hard and nearly impossible to extract the kernel and logits from the pre-trained text classification models because they act like a blackbox where only the output, which is always text, can be considered. Both implemented transformer architectures for emotion detection in audio and video are not suitable for feature combination because of their lack of usable interfaces to extract feature information. These architectures will also only give their output logits. Only the two convolution models can give the required information for kernel learning or other feature combination. As a solution for these prob-

[1] https://github.com/lucidrains/vit-pytorch

lems this thesis will only take a look on scoring methods as a simple form of mapping the three outputs to one final emotion. There are some different methods which are suitable.

Scoring Methods

A scoring method is a way to weigh multiple inputs by their importance for the final output. For example the first architecture is less crucial for the result as the second and third architecture. The task is to find a optimal function $f : \mathbb{V}^{n \times m} \rightarrow \mathbb{V}^n$ where n describes the amount of classifiable emotions and m counts all possible inputs. In this case $n = 7$ for the six base sentiments together with neutral and further $m = 3$ for the three given modals.

Neural Networks which try to learn their own weights are a good design for the simplest approach for scoring without further complexity. Here a single linear layer is used with the predictions of all training data and the actual output. After the learning phase this model should have an internal weight system for the multimodal input. As they try to solve an optimization problem by minimizing the loss they optimize the function f analogous to this. A drawback of this method shows the blackbox-like behavior of neural networks. After training and testing it is impossible to tell which inputs were used for the final result and which were discarded or only had a small impact. Furthermore, the supervised model architecture might also be a drawback in comparison the the following unuspervised approaches.

Brute-Force is a more elaborate method regarding time complexity. The idea is to permute all possible weights for each input. Looking back at $f : \mathbb{V}^{7 \times 3} \rightarrow \mathbb{V}^7$ it can be simplified and adapted to feature a weightened and parameterized function

$$f(a, b, c) : a\mathbb{V}_0^7 \oplus b\mathbb{V}_1^7 \oplus c\mathbb{V}_2^7 \tag{1}$$

where $a, b, c \in [0, 1]$ denote the different weights for each output and $\oplus$ defines the arithmetic operator for combining the vectors. The goal is to minimize the difference between the actual target $\hat{f}$ and the calculated output.

$$\min\Big(\hat{f} - f(a, b, c)\Big) \tag{2}$$

As already mentioned the brute force algorithm will permute all weights with a step size of 0.01. This results in a total of $101^3 = 1,030,301$ permutations. Reducing the step size will increase the amount of possible test cases cubically which underlines the high time complexity. It might also fail at finding the most optimal global solution, only providing the local optimum in the permutation space. To find the global solution the function has to leave the discrete numeric space which is impossible for machines.

Major Voting is a strategy which only works with more than two different inputs. The main idea is quite simple: the most voted emotion must be the corresponding emotion. For example if the input is $(\text{sad}, \text{sad}, \text{neutral})$ the calculated output should be “sad” due to its major occurence in the space. This is also a more simple method and will not solve the optimization problem. Also there is no differentiation between the input models.

```python
def major_vote(L: list[int | float | str]) -> int | float | str:
    d = { } # dictionary for counting the elements
    for i, e in enumerate(L):
        if not e in d:
            d.update({e: 1})
        else:
            d[e] += 1
     # sort by value, highest value at pos 0
     return sorted(d, key= d.get, reverse=True)[0]
```

Listing 4.3.: Python implementation for the major vote algorithm.

This approach will only focus on the output. If two models vote equally for the wrong label the result is also wrong. Thus it is quite efficient regarding complexity and also in comparison to the already proposed scoring methods like brute-force. An implementation for the major voting system is shown in Listing 4.3

There exist many other possible methods. Every clustering algorithm like k-means would give a reliable result by considering the input labels as cluster points and mapping it to natural numbers which then will be observed as emotion. Rule based classification approaches as for example decision trees can also be used. Analyzing all methods would exceed the scope of this theses and might be related to future work.

Architecture

Modal	Architecture	Trainable Parameters
Text	Llama 3	$\approx 4,000,000,000$
	Multi EmoBERT	
	T5	
Audio	Convolution	2,176,474,377
	Audio-Transformer	1,394,574,473
Video	Convolution	2,504,047,929
	3D Vision Transformer	1,266,615,309
Multimodal Scoring	Linear Layer	147

Table 4.4.: Overwie of all trainable parameters for each unimodal architecture.

All unimodal architectures have already been proposed and discussed before in this chapter. Looking back at the abstract overviews for each modal as shown in Figure 4.2, Figure 4.4 and Figure 4.8 some tasks have emerged. First of all due to the text-to-text understanding of the natural language models the output has to get mapped back to the one-hot vector which is

already given by the audio and video based architectures. Secondly for the neural network approach a dataset out of the results from the original datasets is needed. Therefore every model saves its prediction. Table 4.4 shows a final overview over all trainable parameters for each model and also for the multimodal scoring method featuring a neural network. All models are good comparatively regarding their size and amount of trainable weights. The scoring network is the smallest of all thus it includes only a single linear layer which maps three vectors to a single vector.

Chapter 5.

Evaluation

This chapters presents all results for the described and discussed implementations. Therefore, a setup for the evaluation part is shown with a look on accuracy measuring functions especially the precision under recall curve. After this introduction and overview the actual results for the experiments are shown, visualized and further discussed.

5.1. Setup

As already explained, each model is trained on both datasets for 20 epochs and a 90/10 split in training and testing. Each epoch runs through the whole training data which is randomly shuffled every time for better and more efficient training. The preprocessing follows analogously to the implementation presented in the chapter before. The normalization of the data entries is done globally to bring the values in a uniform space between −1 and 1. Every training loop consists of the Adam optimizer and the cross entropy loss function to calculate the gradient. After this each trained model is also tested with each dataset to show if a model trained on CREMA-D is capable to detect emotion in the MELD data and the other way around. The only data which is not tested is the text data from the CREMA dataset. Due to its lack of a variety of sentences this would not be necessary. The dataset consists only of twelve different sentences in around 7000 videos. Every sentence is acted in the different emotions. It is neither learnable nor testable for the further evaluation. After this unimodal detection, all predictions for the test and trainings datasets are saved in an external dataset for the multimodal emotion classification so they do not have to evaluated each time for the three presented methods. To measure the accuracy different metrics are used. Initially as example for the text classification the precision is calculated by simply counting how many of the predictions are actually equal to the expected output. Besides it is counted how many and which emotions are wrongly labeled by the model to display a heatmap and to analyze if there are easy mix-ups between similar feature spaces. This is also repeated for each experiment. For the following sections, the calculation of the accuracy changed to a more informative and meaningful metric.

5.1.1. Accuracy Metrics

Precision under recall describes an accuracy metric to evaluate the quality of classification tasks. It provides three scores for the comparability of the values. For the further overview it is necessary to define some variables to later understand the components of this metric.

- $\boldsymbol{T_P}$ → True Positive: Amount of all correctly classified values.
- $\boldsymbol{T_N}$ → True Negative: Amount of all correctly zero-classified values.
- $\boldsymbol{F_P}$ → False Positive: Amount of all falsly predicted labels
- $\boldsymbol{F_N}$ → False Negative: Amount of all falsly zero-classified labels

Zero-classified are all labels, where the model predicted that these are not the corresponding emotion. Out of these values, the four important metrics can be derived from this. The accuracy shows the proportion of all correctly classified labels including the quantity of all true positive and true negative classifications.

$$\text{Accuracy} = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \tag{1}$$

By looking at the true positive values the precision of the model can be calculated. It describes the proportion of all correct classified values.

$$\text{Precision} = \frac{T_P}{T_P + F_P} \tag{2}$$

In addition, the recall shows how accurate the model predicted the classes by looking at the true positive and false negative values.

$$\text{Recall} = \frac{T_P}{T_P + F_N} \tag{3}$$

At last the F1-score can be calculated to measure the quality of the classification model. It describes the harmonic mean between precision and recall and is denoted by

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

All in all these metrics are normally used for each label instead of for the whole classifier. Thus for a more clear arrangement of the results regarding the amount of different classifier and multimodal combination methods the following results are calculated for all labels in total showing the precision, accuracy, recall and quality of each classifier. Also this global observation of precision under recall helps to ignore imbalances within the datasets. This problem would otherwise occur while evaluating measurements with the MELD dataset. Furthermore the area under recall (AUC) metric describes the area under the precision-recall curve. A perfect model would have an AUC of 1, showing that the precision under recall is also always 1. In turn a completly untrained model has an AUC of 0. In general it shows how high the precision under the amount of all correct classified labels, to see if there are many misclassification, even if the model seems to have a high accuracy.

5.1.2. Experiments

As already discussed there are several experiments to do for an appropriate showcase of the model functionalities. After they got trained they will be evaluated using their normal test

data. Each model will be tested with the MELD data as well as the CREMA data to look if these models are able to understand the data from each other without further training. The best models for each modal will be choosen for the multimodal classification which follows the same testing procedure as before. As presented in the previous chapter, the training for an output fusion with a simple neural network, major voting system and the brute-force approach are simulated and evaluated. The following listing summarizes the following experiments:

- **Experiment 1:** Unimodal emotion detection in **text** with MELD
- **Experiment 2:** Unimodal emotion detection in **audio**, trained on each dataset and tested each permutation.
- **Experiment 3:** Unimodal emotion detection in **video**, trained on each dataset and tested each permutation.
- **Experiment 4** Multimoda emotion detection by **label fusion** with the three presented methods: Neural Network, Brute-Force and Major Voting

5.2. Results

Experiment 1 - Unimodal Results for Classification in Text

	CREMA-D				**MELD**			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
Llama	nA	nA	nA	nA	0.372	0.546	0.372	0.385
Multi-EmoBert	nA	nA	nA	nA	**0.603**	**0.594**	**0.603**	**0.597**
T5	nA	nA	nA	nA	0.559	0.581	0.556	0.559

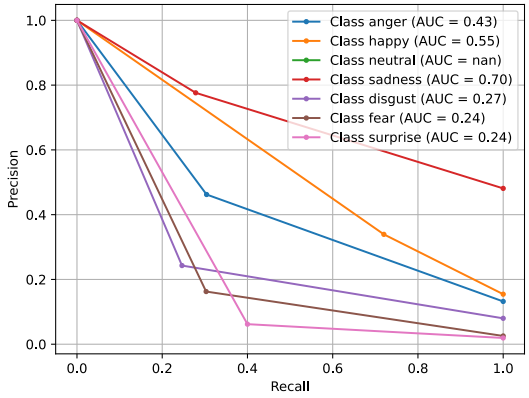
Table 5.1.: Accuracy for unimodal emotion classification in text.

As noted in Table 5.1 it was not possible to measure the accuracy, precision and other metrics for the CREMA-D dataset within the text data. This results from the usage of the same sentence for each emotion. Therefore they change the emotionaly meaning every time so the model is unable to correctly learn any features from the given training set. Fortunately some results are achieved with the MELD dataset, which features a high variety of different sentences. The only prompt-driven model, Llama performes the worst of all three models with only 37% accuracy which still performed better than just guessing ($\frac{1}{7} = 14.3\%$ accuracy) the right emotion. Here the model mostly tried to predict other emotion instead of neutral, leading to a high misclassification rate in every given label which is hightlighted by (b) in Figure 5.1. Llama oftenly predicts happiness wrong which shows that it interprets the sentences most of the time as an happier atmosphere showing the missing understanding of the dataset although the context is presented by the prompt before. Continuing with the Multi-EmoBERT model, this architecture performed the best with a small gap to T5. A further look at Figure 5.1,

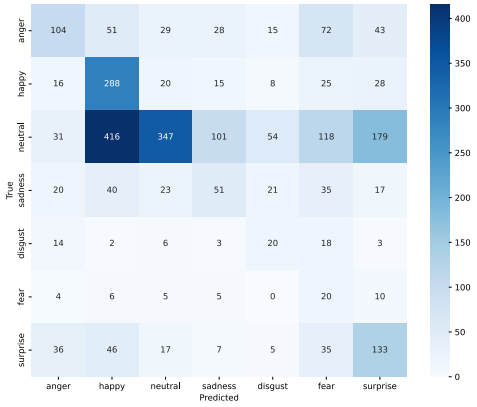
especially at (c) and (d), shows that the neutral emotion was still the best learned emotion and is also the most common emotion in the MELD dataset. However in contrast to Llama, EmoBERT showed in total better results in every category. The precision-recall curve in (c) notes that fear was learned and classified not well at all, but as showcased in the heatmap distribution from (d) it is still a result from the imbalance of the dataset. Here, fear was misclassified the most but there were only around 100 samples in the test dataset while nearly a third was labeled as neutral. All other emotions are aligned as good as fear and neutral to the distribution in the MELD data. It is important to note, that all of the three models, especially EmoBERT are always guessing different emotions even though they will have high accuracy by only classifying the sentences as neutral. This problem will occur again in the video and audio detection. With almost equal results T5 shows mostly similar behavior in training and testing, however the small differences are visible in the emotion with less occurence. In comparison to EmoBERT, nearly every other sentiment was misclassified more often, however, these amounts are also very small. Interestingly, the T5 model was able to label sadness a bit more accurately than every other presented text model which can be an influence of the small yet existing architecture differences and the difference in data for pretraining the weights. The most visible contrast between the two models can be found in the fear and disgust emotion. As shown, the T5 model tries to classify fewer sentences as a further result of the imbalance from MELD data. Llama struggles the most as it is unable to identify the structure of the dataset. In this case it shows the high difference between actual training of the weights and the use of a possible small context memory in these architectures. To look at similarities between the three architectures, both T5 and EmoBERT are able to differentiate perfectly between disgust and fear while it is the only emotion without any misclassification. As these are very rare labels in the given dataset this can be a random occurence.

Overall, it can be observed that text classification models for emotion detection can be a good choice in some cases. Generally these large and public models like Llama3 or ChatGPT, which are seen as the state-of-art tools for nearly every problem, are not able to correctly recognize the correct sentiment even with much context. However, it is currently not publically available how much context these models can handle, store and use, or if most of the given input was mostly ignored by the architecture. In comparison, the directly trained models like EmoBERT and T5, where the user has a chance to fully integrate the context into the given weights, are more suitable for these problems. Nonetheless they are in most cases only able to detect emotion from the specific dataset due to the dependencies within the data. In fact MELD offers only quite specific sentences with many insider words and important context from the series where it finds its origin. While looking at the tokenizer and encoder, there are mostly names from character and specific insider words from this TV-show available. Therefore the model is able to understand the emotional context from this series but it would struggle with for example sentences from a normal conversation. As it turned out later, they will easily overfit while training each model. This could also be the case here but could not be further evaluated due to the lack of necessary data. The approach to detect emotion within the CREMA-D dataset shows that there are no specific rules for a general classification. There are many sentences with different meanings as stated earlier in the background. Every human-made statement and even statements by artifical intelligence can include irony or can be acted differently. It

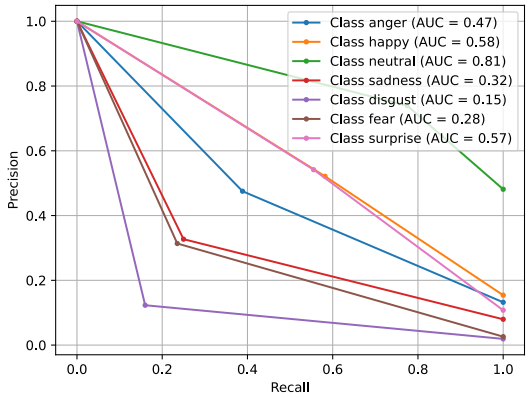
should be nearly impossible to find general patterns to reduce overfitting by a large scale as text-based emotion detection must always look at the context from the full conversation.



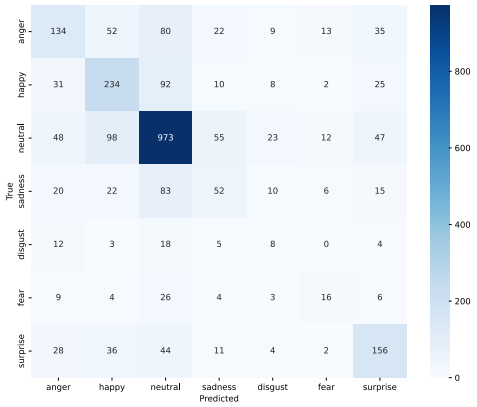
(a) Precision-Recall Curve for Llama3



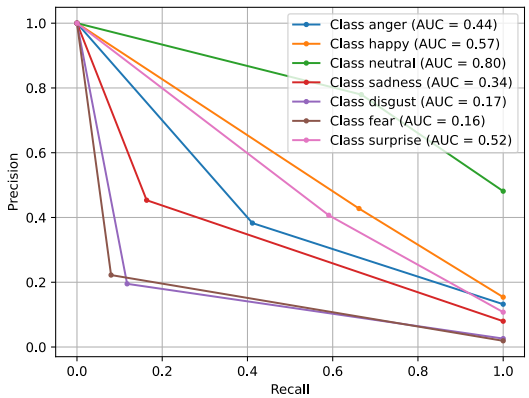
(b) Heatmap classification distribution for Llama3



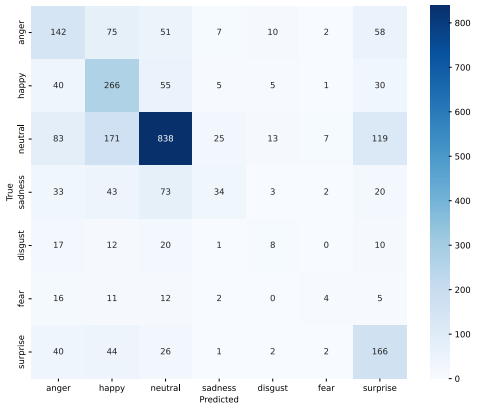
(c) Precision-Recall Curve for EmoBERT



(d) Heatmap classification distribution for EmoBERT



(e) Precision-Recall Curve for T5



(f) Heatmap classification distribution for T5

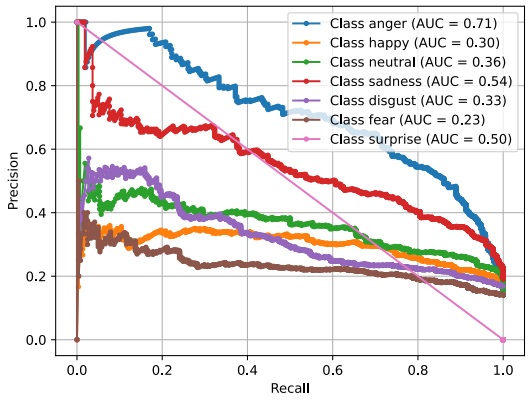
Figure 5.1.: Text-Models: Precision-Recall Curve for each label and heatmap distribution from testing and training with the MELD dataset

Experiment 2 - Unimodal Results for Classification in Audio

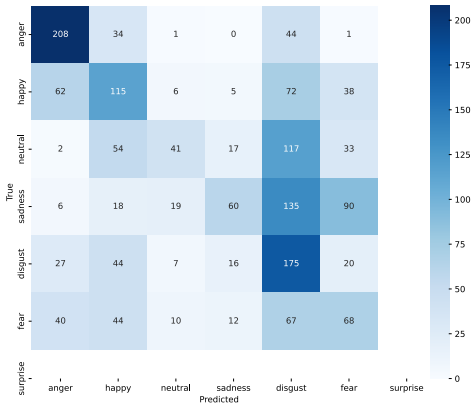
	CREMA-D				MELD			
	Acc.	Precision	Recall	F1	Acc.	Precision	Recall	F1
Conv. (C)	**0.409**	**0.449**	**0.409**	**0.387**	0.143	0.021	0.143	0.036
Transf. (C)	**0.463**	**0.456**	**0.463**	**0.427**	0.151	0.024	0.151	0.039
Conv. (M)	0.115	0.042	0.115	0.046	**0.478**	**0.223**	**0.478**	**0.309**
Transf. (M)	0.134	0.051	0.134	0.055	**0.461**	**0.213**	**0.461**	**0.291**

Table 5.2.: Accuracy for unimodal emotion classification in audio trained on each dataset. (C) = tested with CREMA, (M) = tested with MELD

Table 5.2 summarizes the evaluation results for emotion classification in audio data. The first look shows, that it perfomed not as good as the text models, however they still reach nearly 50% accuracy. Starting with the convolution models, tested on data from the CREMA-D dataset, Figure 5.2 visualizes the performance of the architecture trained on each set. The heatmap in (b) shows that there are many misclassifications within the CREMA-trained model. In addition while looking at (a), it was able to classify the strong emotions like anger and sadness are detected way better than the sentiments which are not as present in voice as expected.



(a) Precision-Recall Curve, trained on CREMA

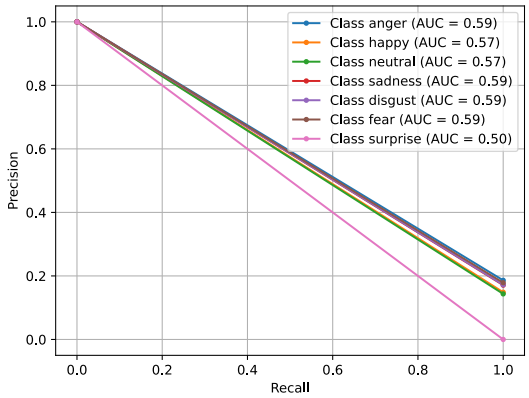


(b) Heatmap classification distribution, trained on CREMA

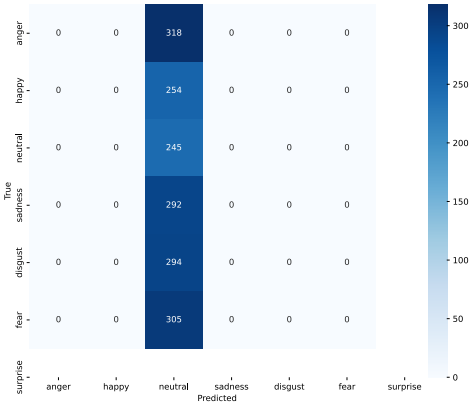
Figure 5.2.: Precision-Recall Curve for each label and their heatmap distribution from training on an audio convolution based network with CREMA-D and tested with the same dataset.

Further the model struggles with the correct labeling of neutral while annotating actual emotions which normally should contains features. In total it achieved 41% accuracy with nearly

45% precision. In comparison, the MELD model struggles with the CREMA data. Here it classifies only neutral for every data entry which is underlined by (a) from Figure 5.3. This problem originates in the actual training with MELD due to the imbalance of the dataset. As shown in (b), the precision-recall curves for each corresponding emotion are overlapping a lot, showing that there were no features analyzed by the architecture. Only surprise is not part of the pattern as it is not featured in the CREMA-D dataset.



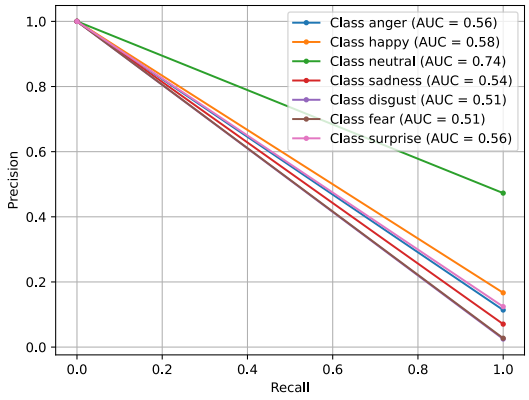
(a) Precision-Recall Curve, trained on MELD



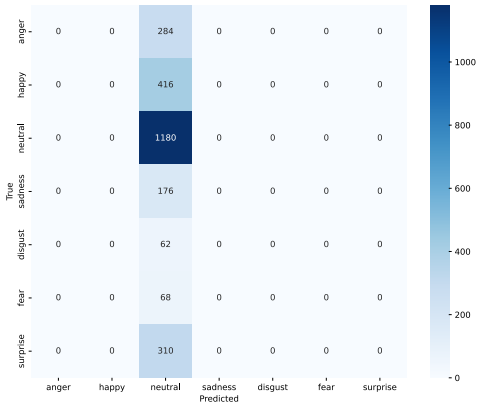
(b) Heatmap classification distribution, trained on MELD

Figure 5.3.: Precision-Recall Curve for each label and their heatmap distribution from training on an audio convolution based network with MELD and tested with the CREMA dataset.

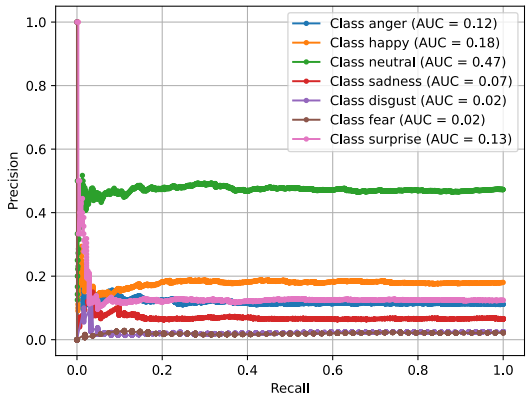
For comparison, the MELD model achieved only 14% accuracy, which is equal to a result which could be obtained by randomly guessing the label. Also the low precision of 2% shows that there was nearly no classification done by the architecture. While testing with the MELD dataset, interesting results can be observerd. As shown in Figure 27, the corresponding model which is trained on the same data, still predicted neutral every time due to the imbalance. This is verified by (a) where neutral has an AUC score of 0.74 instead of the expected 0.5, showing that even the test dataset is imbalanced and contains mostly one label. Therefore, there are less misclassifications and the model achieved an accuracy of 48% but with a low precision of only 22%. This shows, that most of the time a model tries to increase only the accuracy — in general to minimize the loss — without further notice about the actual precision. Meanwhile, the CREMA-D trained model overfitted and nearly only guessed anger as it is the most learned emotion as shown in (d). However (c) shows, that there also are many misclassifications within the testing. There are four emotion which are never classified by the CREMA convolution model but as three of them are quite rare in the MELD dataset, they have a low AUC score. Again neutral is outstanding from the other results as it has no misclassifications. These values are also represented in the low accuracy of 11.5% and a much lower precision of 4.2%. This is even worse than just randomly guessing the correct label, showing that the emotion detection is dataset-specific and not generally applicable.



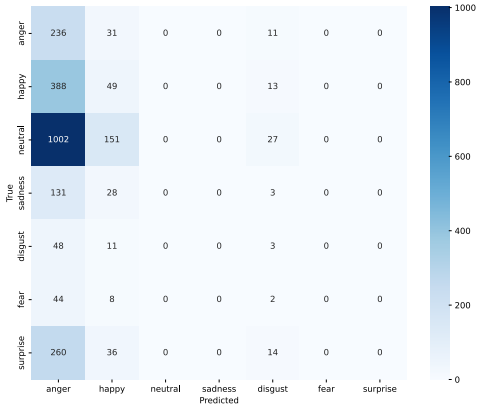
(a) Precision-Recall Curve, trained on MELD



(b) Heatmap classification distribution, trained on MELD



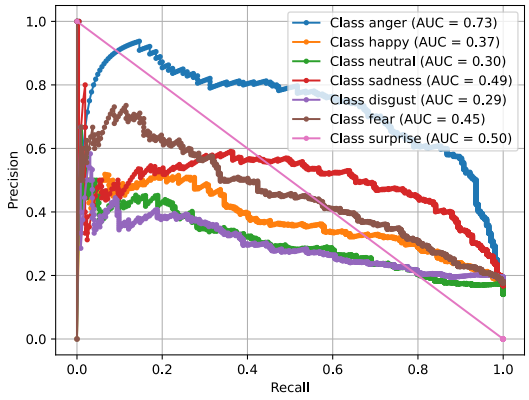
(c) Precision-Recall Curve, trained on CREMA



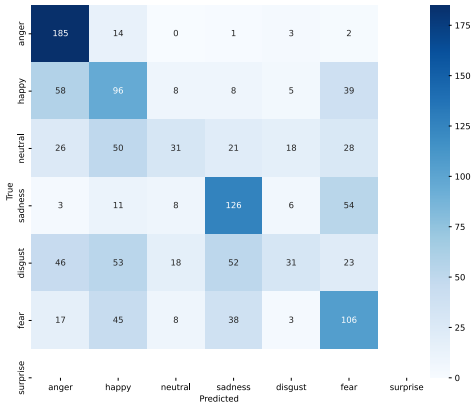
(d) Heatmap classification distribution, trained on CREMA

Figure 5.4.: Precision-Recall Curve for each label and their heatmap distribution from training on an audio convolution based network and tested with the MELD dataset.

The transformer models are showing mostly the same behavior as the convolution moarchitectures but with a small bit of improvement in the scores. First of all, the plots for testing the models with the respective other dataset are similar to before. MELD again just predicts for neutral and CREMA-D overfits and tries to label anger most of the time. Even for the normal testing with the same dataset for each trained transformer, a similar behavior to the convolution models can be observed. As before, CREMA-D was able to recognize various emotion especially anger as it seems to be the clearest sentiment which can be recognized in the voice as shown in Figure 5.5. In general, this model achieved 5% more accuracy and up to 10% increase in precision than the convolution model. The heatmap in (b) shows that there are less misclassifications, which verifies a better training. For MELD it performed the same with around 50% accuracy and the same low precision. This is again a result of the permanent and constant labeling to neutral as shown in Figure 5.6.

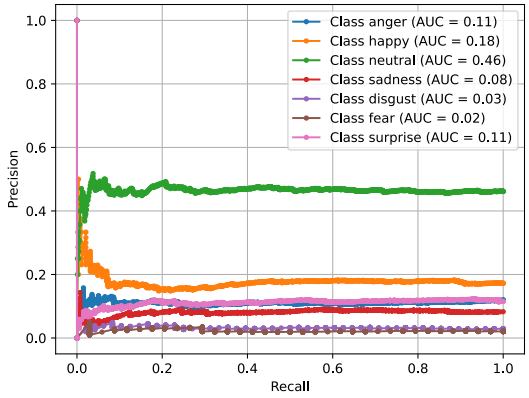


(a) Precision-Recall Curve, trained on CREMA

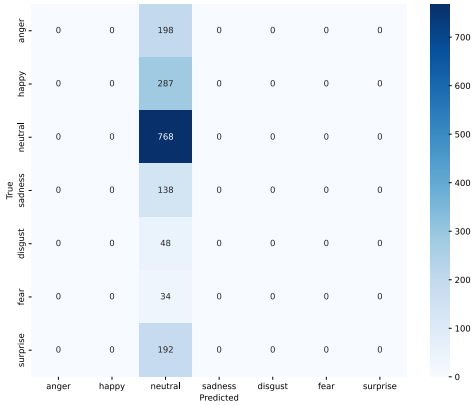


(b) Heatmap classification distribution, trained on CREMA

Figure 5.5.: Precision-Recall Curve for each label and their heatmap distribution from training on an audio transformer based network and tested with the CREMA-D dataset.



(a) Precision-Recall Curve, trained on MELD



(b) Heatmap classification distribution, trained on MELD

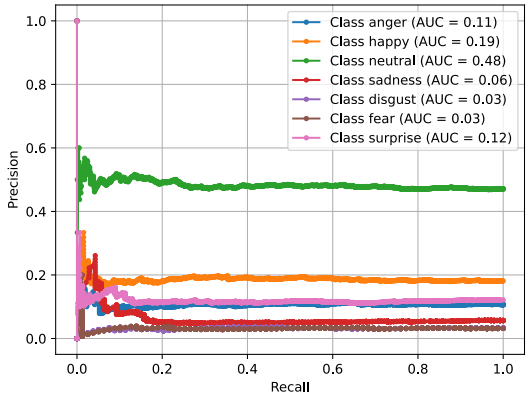
Figure 5.6.: Precision-Recall Curve for each label and their heatmap distribution from training on an audio transformer based network and tested with the MELD dataset.

Experiment 3 - Unimodal Results for Classification in Video

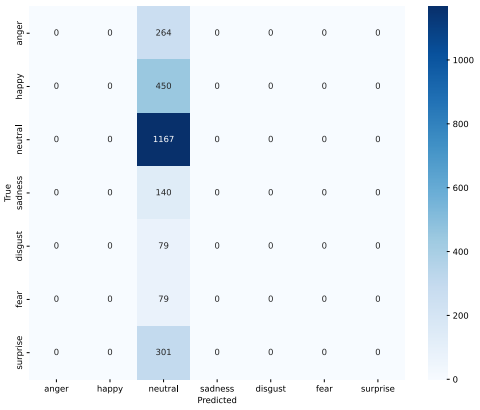
	CREMA-D				MELD			
	Acc.	Precision	Recall	F1	Acc.	Precision	Recall	F1
Conv. (C)	**0.749**	**0.747**	**0.747**	**0.742**	0.154	0.024	0.154	0.041
Transf. (C)	**0.544**	**0.577**	**0.544**	**0.539**	0.146	0.021	0.146	0.037
Conv. (M)	0.295	0.049	0.295	0.043	**0.473**	**0.223**	**0.473**	**0.303**
Transf. (M)	0.219	0.057	0.219	0.057	**0.472**	**0.223**	**0.472**	**0.303**

Table 5.3.: Accuracy for unimodal emotion classification in video trained on each dataset. (C) = tested with CREMA, (M) = tested with MELD

Looking at the results for classification in audio, these results are quite similar. However, there was one architecture which performed with great accuracy. Visible in the following Table 5.3 the models are still struggeling with the imbalance of the MELD dataset. As for audio classification it was the most efficient way to only predict the neutral emotion while always achieving a high accuracy of around 47%. However the precision of only 20% and the low F1-score are still showing this problem underlined by the visualization in Figure 5.7 and Figure 5.8. Just as they look similar to each other, they are also alike to the figures from the training of all audio models with MELD. All other emotion except neutral are untrained thus the model never learned features from the video input and only trained to guess neutral.



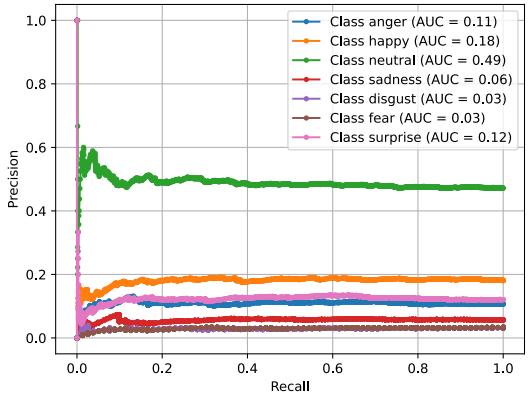
(a) Precision-Recall Curve, trained on MELD.



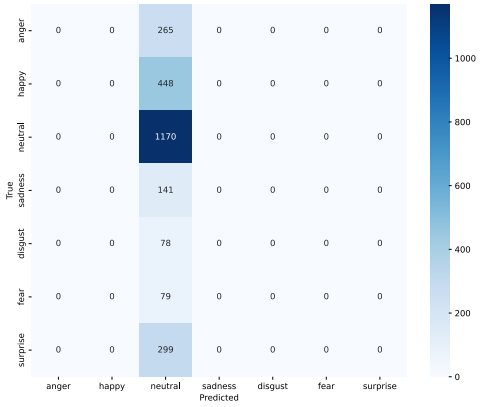
(b) Heatmap classification distribution, trained on MELD.

Figure 5.7.: Precision-Recall Curve for each label and their heatmap distribution from training on an video convolution based network and tested with the MELD dataset.

Also similar to emotion detection in audio, the presented models are still overfitting due to the lack of uniform datasets. Even though the preprocessing tried to bring the videos to a similar level with the active speaker detection and cropping, there are still many artefacts. The TV series character from MELD features many movements from the head of the acting person which is not present in the CREMA data where the face is always looking directly straight into the camera. These effects can also be seen in Table 5.3. As in audio, the precision for each model tested with the data set with which it was not trained is really low with under 6%. Although the accuracy for the MELD trained models reaches the value, which is expected for guessing one of the seven emotion. For this reason the plots are not shown for the cross-testing as they will look the same as (c) and (d) from Figure 5.7 or any figure from audio detection.



(a) Precision-Recall Curve, trained on MELD.



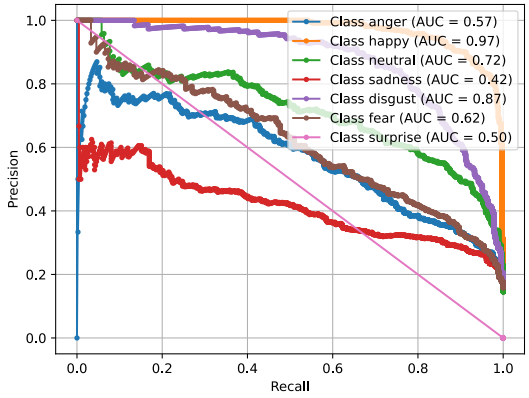
(b) Heatmap classification distribution, trained on MELD.

Figure 5.8.: Precision-Recall Curve for each label and their heatmap distribution from training on an video transformer based network and tested with the MELD dataset.

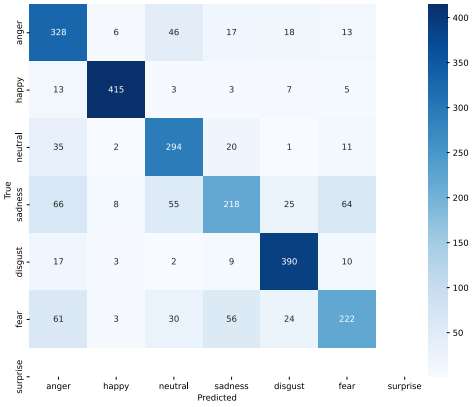
The higher results from the CREMA-D trained models are more interesting. Again the overview in Table 5.3 shows that the convolution architecture achieved an accuracy of nearly 75% which is the highest score achieved by any model so far. Figure 5.9 shows in (b), that the diagonal with the correct classified emotion is highlighted. This correlates with a good classification. There are still some misclassification, mostly for anger and fear as well as fear with sadness and anger with sadness which are sharing many facial features. As before, the CREMA dataset did not include any samples with surprise so there were no evaluation for this label possible.

The graphical representation of the precision-recall curve in (a) from Figure 5.9 desribes that the model performed best with happy and closely followed by disgust. Due to the overacting by the people which are shown in the video samples, both emotions are cleary differentiable from each other. Followed by this explanation that the model looks mostly at the strong shown features from the faces, neutral performed much worse than the two other well learned emotions due to this sentiment should not cover any features. Furthermore, sadness performed the worst out of all of the six labels due to multiple misclassifications with fear and anger as they differ mostly in one or two features. Corresponding to Ekmans Facs system which was explained before, all three emotion share all information around the eyes and they are only

differentiable by the lower part of the face. The convolution model might lay its focus more on the upper half which can lead to this kind of training and testing behavior.



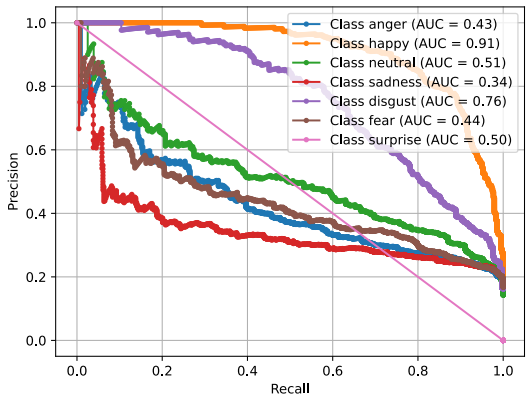
(a) Precision-Recall Curve for CREMA



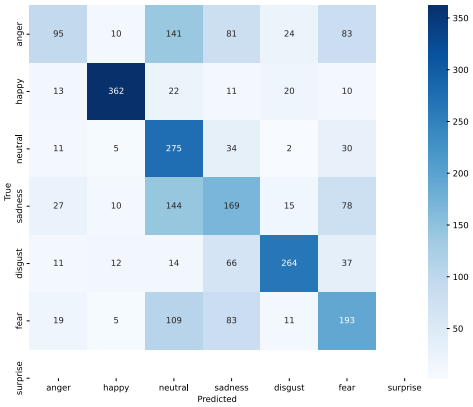
(b) Heatmap classification distribution for CREMA

Figure 5.9.: Precision-Recall Curve and their heatmap distribution from training on an video convolution based network and tested with the CREMA-D dataset.

Nearly the same observations can be found within the transformer model. Visible in (b) from Figure 5.10 the diagonal line with the correct classified labels is still existing and clearly recognizable. However there are also quite a lot of misclassified outliers, mostly where the actual label is anger, sadness and fear. This problem follows the explanation about the convolution model but is more visible. It shows that the model did not learn the correct features as fast as the convolution model which can be a result due to the difference in the amount of trainable parameters. With more time in training which was not possible due to the limitations in this thesis, the model could achieve the same results as from the convolution model.



(a) Precision-Recall Curve for CREMA



(b) Heatmap classification distribution for CREMA

Figure 5.10.: Precision-Recall Curve and their heatmap distribution from training on an video transformer based network and tested with the CREMA-D dataset.

Looking at (a) from Figure 5.10 it is still clear that happiness was well learned, followed by disgust. The other emotions show that there is more proper training missing, to differentiate between the smaller yet important features.

In total, there are some facial features which can be learned by neural networks or similar deep learning architectures. However, there are still many problems with the existing datasets as there is no standard or good comparability between each other. This leads to the described overfitting effect which was found in every modal yet since there are different artefacts like shaking heads or other disturbances. With better active speaker recognition, artifical camera stabilization or interpolation to clean up bad frames there are many possibilities to improve and generalize these models within video based emotion detection.

Experiment 4 - Multimodal Results

For multimodal evaluation the best model of each modal was picked, as it is not necessary to test the same experiment with less precise results but identically trained models. For MELD the EmoBERT model, the audio convolution and the video convolution architecture are chosen. As there are no metrics and trained text models for CREMA-D, only the audio transformer and video convolution models are used for the multimodal evaluation. To simulate the missing results from text detection a random label is picked for each sentence. Without a third modal the major vote strategy would not work properly. Table 5.4 presents the results of the multimodal experiment for each dataset. In general it can be observed that none of the three label combination methods brought any optimization or any higher accuracy or precision. Starting with CREMA-D, the major vote algorithm performed not as good as expected. With only 25% accuracy it is still better than random voting for a label. Though, looking at the voting options for each label there are many disagreements within the three models.

	CREMA-D				**MELD**			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
Neural Network	**0.759**	**0.758**	**0.756**	**0.751**	0.470	0.066	0.142	0.090
Brute Force	0.735	0.737	0.736	0.730	**0.603**	**0.594**	**0.603**	**0.597**
Major Voting	0.252	0.273	0.250	0.243	0.471	0.067	0.142	0.092

Table 5.4.: Accuracy for multimodal emotion classification.

As the audio and video convolution architectures are having a big gap in accuracy of around 30%, there are many labels which are misclassified, leading to the bad voting. In addition to the random annotions to simulate the text classification, there might be a bias where the majority votes for a false positive result. The brute-force algorithm shows, that the weights are mostly given to the video convolution model as it has the highest accuracy out of

all three modals. Further, it shows that there are little to no supplementary labels in each unimodal prediction. Therefore, it is the most accurate to do the prediction only with the video model results. The neural network achieved a higher accuracy of 1-2% in total, which is nevertheless very low. It can be the case that the model learned to predict emotions like neutral whenever all three modals are contradicting to each other. However this is also just a rare case or the model is predicting to a not neutral label whenever this case occurs.

In contrast to the CREMA-D results, MELD is following most of the behavior of the unimodal evaluation. The neural network is still always predicting the neutral emotion. As before, it achieves an accuracy of around 50% due to the imbalance of the dataset. Followed by this, major voting also only results in neutral. Two of the tree models are continuously calling for the same emotion. Only the text model was able to predict a variety of emotion. Here the brute-force approach also puts all weights on this single modal, as it was the case with CREMA-D. Therefore the results are mostly similar to these from the emotion detection in text with the EmoBERT architecture.

Generally these three multimodal combination are not optimal for this case. Label fusion is a hard task and generally profits from models where different labels are classified true positive and complement each other. It can profit from architectures like mixture of experts, where the same model is trained multiple times in parallel to possible learn different features every time. With a higher amount of labels for major voting or the neural network, better decision could be drawn. Looking back at the related work, Vielzeuf et al. [62] used weighted mean as a kind of label combination and also only achieved an accuracy of 50%. In comparison to the other presented works, they achieved higher accuracy with feature combination within the architectures. Here it will be part of future work while this technique will require more access to different model layers. As the three modals are acting more like a black box with finetuned output heads, it exceeds the scope of this thesis due to the need of building an accessable clone of these architectures and pretrain them first under the same conditions.

Chapter 6.

Conclusion

This chapter will give an overview over the findings of this thesis and which limitations occured during the implementation and evaluation. Further, an outlook about optimization ideas and other future work about this research topic, is presented.

6.1. Summary

The background of this thesis discussed mutliple approaches for emotion detection in the three given modals and where different features can be obtained. There are a lot of supervised and also unsupervised architectures which could lead to promising results. Mostly theses ideas are based on neural network architectures — convolution, recurrent and transformer models — or based on similar machine learning algorithms. Further both datasets, CREMA-D and MELD are introduced: CREMA-D is a scientific dataset with a static view angle on the actors face. The drawback stems from the usage of only twelve different sentences, distributed in 7,000 videos. MELD is extracted from the TV-show "Friends" and shows short clips with multi-conversation input and many other challenges. Here the drawback origins in a high imbalance within the labels.

Related work showed that many approaches with neural networks and machine learning are already implemented. These are also commonly based on convolution or variations of recurrent networks. Only any implementations with transformer architectures could not be found yet. Furthermore, the contributions are explaining multiple approaches for feature combination to achieve high accuracy in multimodal emotion detection.

The own implementation starts with three pre-trained text models: Llama 3, Multi-EmoBERT and T5. The chapter explained that it is necessary to feed the context for each dataset in to these architectures. Llama accepts only some way of prompt engineering as a chat like option. EmoBERT and T5 are offering an interface for further training of the weights. Additionally, the feature extraction method for emotion understanding in audio was discussed. OpenSmile offers an extraction of many details and metrics about the audio, based on fourier transformation. Therefore, information like loudness or voice intensity were extracted. For both, audio and video detection, the convolution and transformer architecture are shown. Additionally for video, further preprocessing was implemented where the videos are cropped down on to the active speaker. This was necessary due to the different video and camera formats from each dataset.

To summarize the results: artifical intelligence is able to recognize emotions in different modals. Even the state-of-the-art transformer models, which were introduced recently, are able to keep pace with the traditional convolution architectures. However, there are many challenges which have to be handled first and some general issues, which are hard to fix. The first challange starts with the understanding of famous models like ChatGPT or Llama, which are used every day from many people. This thesis shows that there is much context necessary for more precise predictions, which is only conditionally feasible. These contextual dependencies also exist for every other model, so it is nearly impossible to build an architecture which is able to classify emotion in every scenario. In this case overfitting within each architecture was observed in every evaluation step. Emotions can be detected but the theories by Ekman or other psychologists are not universal rules. At the end these patterns are different for every person, sometimes more and sometimes less but relie on the foundation of these theories.

Another finding is the interesting fact, that every modal is important for a multimodal evaluation. As stated in the motivation, there might be at least one modal which could not deliver important information to the multimodal result. The evaluation shows, that there are always usable features within each modal which results in more than 50% accuracy. Even though audio classification did not add any benefits to the label combination, it might improve a feature combination method.

In comparison to the presented related works, the models performed quite well with up to over 70% accuracy, although this should be considered with caution due to the missing comparability within the used datasets and the presented model architectures.

6.2. Limitations

The biggest limitation is added by the datasets themself. Mostly MELD shows that there are no unified standards for these kind of multimodal collections. There are high imbalances in the data distribution and no unified camera positions, audio recording types and other points to make them comparable to each other. Further limitation also stem from time constraints. Normally these large models will train up to several months for better usability and reliable results. However the scope of this thesis only allowed to train on short time although the results are still representing the foundings as presented before.

6.3. Future Work

As presented by Picard in his book “Affective Computing” [2] the research about emotion detection is still a big topic with many interesting optimization approaches. There exist much more modals than just text, video and audio. Nowadays wearables like smart watches, smart glasses or other devices are able to precisely measure information like blood pressure, heart rate or even directly calculate a stress level. These features could also be used to detect emotion, as for example a higher or lower pulse rate might stand for calm and aggressive emotions. Also the raw environment around the person, like the weather or the atmosphere brought by the surroundings could be used as valuable features.

Furthermore, experiments with feature combination could show better results and offer a high variety of optimization and different approaches. As explained before there are more insights in the given model architectures are needed to provide an interface to early combine necessary information.

To avoid further limitations with the datasets, defining standards and important needs for better comparability between emotion detection approaches is needed. The challenges occur when finding as many data to satisfy large architectures and when finding valuable annotations for supervised learning. This can be done first with unsupervised learning but still needs manual correction by at least one person. However even for humans, it is hard to find the one correct emotion for each data entry.

At last, a just-in-time emotion detection offers high potential for application in job interviews or better interaction with non-human devices. Therefore, further improvements in computation time and complexity of the architectures are necessary.

Bibliography

[1] W. James, "What is an Emotion?," *Mind*, vol. 9, no. 34, pp. 188–205, 1884, Accessed: Mar. 11, 2024. [Online]. Available: http://www.jstor.org/stable/2246769

[2] R. W. Picard, "Affective computing." MIT Press, Cambridge, Mass. [u.a.], 1997.

[3] K. R. Scherer and P. Ekman, Eds., *Approaches to Emotion*, 1st ed. Psychology Press, 1984.

[4] P. Kumar and M. Vardhan, "PWEBSA: Twitter sentiment analysis by combining Plutchik wheel of emotion and word embedding," *International Journal of Information Technology*, vol. 14, no. 1, pp. 69–77, Feb. 2022.

[5] D. Chafale and A. Pimpalkar, "Review on Developing Corpora for Sentiment Analysis Using Plutchik's Wheel of Emotions with Fuzzy Logic," *International Journal of Computer Sciences and Engineering (IJCSE)*, vol. 2, pp. 14–18, 2014.

[6] M. Müller, *Fundamentals of music processing: Audio, analysis, algorithms, applications*, vol. 5. Springer, 2015.

[7] I. R. Murray and J. L. Arnott, "Applying an analysis of acted vocal emotions to improve the simulation of synthetic speech," *Computer Speech & Language*, vol. 22, no. 2, pp. 107–129, 2008, [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0885230807000393

[8] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, "Crema-d: Crowd-sourced emotional multimodal actors dataset," *IEEE transactions on affective computing*, vol. 5, no. 4, pp. 377–390, 2014.

[9] E. Friesen and P. Ekman, "Facial action coding system: a technique for the measurement of facial movement," *Palo Alto*, vol. 3, no. 2, p. 5–6, 1978.

[10] J. C. Hager, P. Ekman, and W. V. Friesen, "Facial action coding system," *Salt Lake City, UT: A Human Face*, p. 8–9, 2002.

[11] B. Wang, A. Wang, F. Chen, Y. Wang, and C.-C. J. Kuo, "Evaluating word embedding models: methods and experimental results," *APSIPA Transactions on Signal and Information Processing*, vol. 8, p. 19–20, 2019, doi: 10.1017/ATSIP.2019.12.

[12] D. Karani, "Introduction to word embedding and Word2Vec," *Towards Data Science*, vol. 1, 2018.

[13] Y. Li and T. Yang, "Word Embedding for Understanding Natural Language: A Survey," in *Guide to Big Data Applications*, S. Srinivasan, Ed., Cham: Springer International Publishing, 2018, pp. 83–104.

[14] K. Choudhary and R. Beniwal, "Xplore Word Embedding Using CBOW Model and Skip-Gram Model," in *2021 7th International Conference on Signal Processing and Communication (ICSC)*, 2021, pp. 267–270. doi: 10.1109/ICSC53193.2021.9673321.

[15] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," *arXiv preprint arXiv:1301.3781*, 2013.

[16] X. Rong, "Word2Vec parameter learning explained," arXiv, 1411.2738, 2016. [Online]. Available: https://arxiv.org/abs/1411.2738

[17] L. Ma and Y. Zhang, "Using Word2Vec to process big text data," in *2015 IEEE International Conference on Big Data (Big Data)*, 2015, pp. 2895–2897. doi: 10.1109/BigData.2015.7364114.

[18] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," arXiv, 1810.04805, 2019. [Online]. Available: https://arxiv.org/abs/1810.04805

[19] E. Fredkin, "Trie memory," *Commun. ACM*, vol. 3, no. 9, pp. 490–499, Sep. 1960, doi: 10.1145/367390.367400.

[20] X. Song, A. Salcianu, Y. Song, D. Dopson, and D. Zhou, "Fast WordPiece Tokenization," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, Eds., Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 2089–2103. doi: 10.18653/v1/2021.emnlp-main.160.

[21] C. Fellbaum and K. Brown, "Encyclopedia of language and linguistics." Elsevier Oxford, 2005.

[22] B. Liu, "Text sentiment analysis based on CBOW model and deep learning in big data environment," *Journal of ambient intelligence and humanized computing*, vol. 11, no. 2, pp. 451–458, 2020.

[23] E. S. Lubana, R. Dick, and H. Tanaka, "Beyond BatchNorm: Towards a Unified Understanding of Normalization in Deep Learning," in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., Curran Associates, Inc., 2021, pp. 4778–4791. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2021/file/2578eb9cdf020730f77793e8b58e165a-Paper.pdf

[24] L. Medsker and L. C. Jain, *Recurrent neural networks: design and applications*. CRC press, 1999.

[25] A. Graves, "Long Short-Term Memory," in *Supervised Sequence Labelling with Recurrent Neural Networks*, Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 37–45. doi: 10.1007/978-3-642-24797-2_4.

[26] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling," arXiv, 2014. [Online]. Available: https://arxiv.org/abs/1412.3555

[27] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention Is All You Need," arXiv, 1706.03762, 2023. [Online]. Available: https://arxiv.org/abs/1706.03762

[28] OpenAI, J. Achiam, S. Adler, S. Agarwal, L. Ahmad, *et al.*, "GPT-4 Technical Report," arXiv, 2303.08774, 2024. [Online]. Available: https://arxiv.org/abs/2303.08774

[29] S. L. Smith, A. Brock, L. Berrada, and S. De, "ConvNets Match Vision Transformers at Scale," arXiv, 2310.16764, 2023. [Online]. Available: https://arxiv.org/abs/2310.16764

[30] R. Sutton, "The bitter lesson, 2019," *URL http://www. incompleteideas. net/IncIdeas/BitterLesson. html*, 2019.

[31] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei, "Scaling Laws for Neural Language Models," arXiv, 2001.08361, 2020. [Online]. Available: https://arxiv.org/abs/2001.08361

[32] S. Deerwester, S. T. Dumais, G. W. Furnas, T. K. Landauer, and R. Harshman, "Indexing by latent semantic analysis," *Journal of the American Society for Information Science*, vol. 41, no. 6, pp. 391–407, 1990, doi: https://doi.org/10.1002/(SICI)1097-4571(199009)41:6<391::AID-ASI1>3.0.CO;2-9.

[33] V. Klema and A. Laub, "The singular value decomposition: Its computation and some applications," *IEEE Transactions on Automatic Control*, vol. 25, no. 2, pp. 164–176, 1980, doi: 10.1109/TAC.1980.1102314.

[34] A. Agrawal and A. An, "Unsupervised emotion detection from text using semantic and syntactic relations," in *2012 IEEE/WIC/ACM International Conferences on Web Intelligence and Intelligent Agent Technology*, 2012, pp. 346–353.

[35] S. D. Gollapalli, P. Rozenshtein, and S. K. Ng, "ESTeR: Combining word co-occurrences and word associations for unsupervised emotion detection," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 1043–1056.

[36] Y.-c. Park, J.-G. Jang, and U. Kang, "Fast Partial Fourier Transform," arXiv, 2008.12559, 2020. [Online]. Available: https://arxiv.org/abs/2008.12559

[37] L. Durak and O. Arikan, "Short-time Fourier transform: two fundamental properties and an optimal implementation," *IEEE Transactions on Signal Processing*, vol. 51, no. 5, pp. 1231–1242, 2003.

[38] L. Wang, P. Luc, Y. Wu, A. Recasens, L. Smaira, A. Brock, A. Jaegle, J.-B. Alayrac, S. Dieleman, J. Carreira, and A. van den Oord, "Towards Learning Universal Audio Representations," arXiv, 2111.12124, 2022. [Online]. Available: https://arxiv.org/abs/2111.12124

[39] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. D. Plumbley, "PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition," arXiv, 1912.10211, 2020. [Online]. Available: https://arxiv.org/abs/1912.10211

[40] Y. M. Costa, L. S. Oliveira, and C. N. Silla, "An evaluation of Convolutional Neural Networks for music classification using spectrograms," *Applied Soft Computing*, vol. 52, pp. 28–38, 2017, doi: https://doi.org/10.1016/j.asoc.2016.12.024.

[41] M. French and R. Handy, "Spectrograms: Turning Signals into Pictures," *Journal of Engineering Technology*, vol. 24, pp. 32–35, 2007.

[42] P. Verma and J. Berger, "Audio Transformers:Transformer Architectures For Large Scale Audio Understanding. Adieu Convolutions," arXiv, 2105.00335, 2021. [Online]. Available: https://arxiv.org/abs/2105.00335

[43] J. Wülfing and M. A. Riedmiller, "Unsupervised Learning of Local Features for Music Classification.," in *ISMIR*, 2012, pp. 139–144.

[44] Y.-C. Lu, C.-W. Wu, C.-T. Lu, and A. Lerch, "An Unsupervised Approach to Anomaly Detection in Music Datasets," in *Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval*, in SIGIR '16. Pisa, Italy: Association for Computing Machinery, 2016, pp. 749–752. doi: 10.1145/2911451.2914700.

[45] M. Jogin, Mohana, M. S. Madhulika, G. D. Divya, R. K. Meghana, and S. Apoorva, "Feature Extraction using Convolution Neural Networks (CNN) and Deep Learning," in *2018 3rd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, 2018, pp. 2319–2323. doi: 10.1109/RTEICT42901.2018.9012507.

[46] J. Selva, A. S. Johansen, S. Escalera, K. Nasrollahi, T. B. Moeslund, and A. Clapés, "Video Transformers: A Survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 11, pp. 12922–12943, 2023, doi: 10.1109/TPAMI.2023.3243465.

[47] F. Krause, *Video Emotion Classification utilizing Temporal Difference Images*. Master Thesis, Otto-von-Guericke University Magdeburg, 2023.

[48] A. Kumar, A. Kumar, V. Vineet, and Y. S. Rawat, "A Large-Scale Analysis on Self-Supervised Video Representation Learning," 2023, [Online]. Available: https://arxiv.org/abs/2306.06010

[49] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lučić, and C. Schmid, "ViViT: A Video Vision Transformer," arXiv, 2021. [Online]. Available: https://arxiv.org/abs/2103.15691

[50] H. Akbari, L. Yuan, R. Qian, W.-H. Chuang, S.-F. Chang, Y. Cui, and B. Gong, "VATT: Transformers for Multimodal Self-Supervised Learning from Raw Video, Audio and Text," *ArXiv*, 2021, [Online]. Available: https://api.semanticscholar.org/CorpusID:233346984

[51] G. Patil and P. Suja, "Emotion recognition from 3D videos using optical flow method," in *2017 International Conference On Smart Technologies For Smart Nation (SmartTechCon)*, 2017, pp. 825–829. doi: 10.1109/SmartTechCon.2017.8358488.

[52] J. Haddad, O. Lezoray, and P. Hamel, "3D-CNN for Facial Emotion Recognition in Videos," in *Advances in Visual Computing*, G. Bebis, Z. Yin, E. Kim, J. Bender, K. Subr, B. C. Kwon,

J. Zhao, D. Kalkofen, and G. Baciu, Eds., Cham: Springer International Publishing, 2020, pp. 298–309.

[53] L. Zhang, Y. Tong, and Q. Ji, "Active Image Labeling and Its Application to Facial Action Labeling," in *Computer Vision – ECCV 2008*, D. Forsyth, P. Torr, and A. Zisserman, Eds., Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, pp. 706–719.

[54] Y. Tong, W. Liao, and Q. Ji, "Facial Action Unit Recognition by Exploiting Their Dynamic and Semantic Relationships," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 29, no. 10, pp. 1683–1699, 2007, doi: 10.1109/TPAMI.2007.1094.

[55] B. Martinez, M. F. Valstar, B. Jiang, and M. Pantic, "Automatic Analysis of Facial Actions: A Survey," *IEEE Transactions on Affective Computing*, vol. 10, no. 3, pp. 325–347, 2019, doi: 10.1109/TAFFC.2017.2731763.

[56] M. Valstar, M. Pantic, and I. Patras, "Motion history for facial action detection in video," in *2004 IEEE International Conference on Systems, Man and Cybernetics (IEEE Cat. No.04CH37583)*, 2004, pp. 635–640. doi: 10.1109/ICSMC.2004.1398371.

[57] H. Shimodaira, K.-i. Noma, M. Nakai, and S. Sagayama, "Dynamic Time-Alignment Kernel in Support Vector Machine," in *Advances in Neural Information Processing Systems*, T. Dietterich, S. Becker, and Z. Ghahramani, Eds., MIT Press, 2001, p. . [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2001/file/a869ccbcbd9568808b8497e28275c7c8-Paper.pdf

[58] F. Zhou, F. De la Torre, and J. F. Cohn, "Unsupervised discovery of facial events," in *2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2010, pp. 2574–2581. doi: 10.1109/CVPR.2010.5539966.

[59] T. Warren Liao, "Clustering of time series data—a survey," *Pattern Recognition*, vol. 38, no. 11, pp. 1857–1874, 2005, doi: https://doi.org/10.1016/j.patcog.2005.01.025.

[60] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "Meld: A multimodal multi-party dataset for emotion recognition in conversations," *arXiv preprint arXiv:1810.02508*, 2018.

[61] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J. N. Chang, S. Lee, and S. S. Narayanan, "IEMOCAP: Interactive emotional dyadic motion capture database," *Language resources and evaluation*, vol. 42, pp. 335–359, 2008.

[62] V. Vielzeuf, S. Pateux, and F. Jurie, "Temporal multimodal fusion for video emotion classification in the wild," in *Proceedings of the 19th ACM International Conference on Multimodal Interaction*, in ICMI '17. Glasgow, UK: Association for Computing Machinery, 2017, pp. 569–576. doi: 10.1145/3136755.3143011.

[63] A. Dhall, R. Goecke, S. Lucey, and T. Gedeon, "Collecting Large, Richly Annotated Facial-Expression Databases from Movies," *IEEE MultiMedia*, vol. 19, no. 3, pp. 34–41, 2012, doi: 10.1109/MMUL.2012.26.

[64] A. S. Patwardhan, "Multimodal mixed emotion detection," in *2017 2nd International Conference on Communication and Electronics Systems (ICCES)*, 2017, pp. 139–143. doi: 10.1109/CESYS.2017.8321250.

[65] M. G. Huddar, S. S. Sannakki, and V. S. Rajpurohit, "Multi-level feature optimization and multimodal contextual fusion for sentiment analysis and emotion classification," *Computational Intelligence*, vol. 36, no. 2, pp. 861–881, 2020, doi: https://doi.org/10.1111/coin.12274.

[66] C. Yang, F. Chen, and S. Tian, "Grouped Genetic Algorithm Based Optimal Tests Selection for System with Multiple Operation Modes," *Journal of Electronic Testing*, vol. 33, p. , 2017, doi: 10.1007/s10836-017-5672-y.

[67] N. Majumder, D. Hazarika, A. Gelbukh, E. Cambria, and S. Poria, "Multimodal sentiment analysis using hierarchical fusion with context modeling," *Knowledge-Based Systems*, vol. 161, pp. 124–133, 2018, doi: https://doi.org/10.1016/j.knosys.2018.07.041.

[68] H. Pham, T. Manzini, P. P. Liang, and B. Poczós, "Seq2Seq2Sentiment: Multimodal Sequence to Sequence Models for Sentiment Analysis," in *Proceedings of Grand Challenge and Workshop on Human Multimodal Language (Challenge-HML)*, A. Zadeh, P. P. Liang, L.-P. Morency, S. Poria, E. Cambria, and S. Scherer, Eds., Melbourne, Australia: Association for Computational Linguistics, Jul. 2018, pp. 53–63. doi: 10.18653/v1/W18-3308.

[69] J. Pennington, R. Socher, and C. D. Manning, "Glove: Global vectors for word representation," in *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, 2014, pp. 1532–1543.

[70] G. Degottex, J. Kane, T. Drugman, T. Raitio, and S. Scherer, "COVAREP — A collaborative voice analysis repository for speech technologies," in *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2014, pp. 960–964. doi: 10.1109/ICASSP.2014.6853739.

[71] A. Zadeh, M. Chen, S. Poria, E. Cambria, and L.-P. Morency, "Tensor Fusion Network for Multimodal Sentiment Analysis," arXiv, 1707.07250, 2017. [Online]. Available: https://arxiv.org/abs/1707.07250

[72] G. Xu, W. Li, and J. Liu, "A social emotion classification approach using multi-model fusion," *Future Generation Computer Systems*, vol. 102, pp. 347–356, 2020, doi: https://doi.org/10.1016/j.future.2019.07.007.

[73] S. Sonnenburg, G. Rätsch, C. Schäfer, and B. Schölkopf, "Large scale multiple kernel learning," *The Journal of Machine Learning Research*, vol. 7, pp. 1531–1565, 2006.

[74] Meta, "Introducing Meta Llama 3: The most capable openly available LLM to date. ai.meta.com." blog entry, 2023.

[75] J. Li and L. Xiao, "Multi-emotion Recognition Using Multi-EmoBERT and Emotion Analysis in Fake News," in *Proceedings of the 15th ACM Web Science Conference 2023*, 2023, pp. 128–135.

[76] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer," arXiv, 1910.10683, 2023. [Online]. Available: https://arxiv.org/abs/1910.10683

[77] B. Hammarberg, B. Fritzell, J. Gaufin, J. Sundberg, and L. Wedin, "Perceptual and acoustic correlates of abnormal voice qualities," *Acta oto-laryngologica*, vol. 90, no. 1–6, pp. 441–451, 1980.

[78] M. Berg, M. Fuchs, K. Wirkner, M. Loeffler, C. Engel, and T. Berger, "The Speaking Voice in the General Population: Normative Data and Associations to Sociodemographic and Lifestyle Factors," *Journal of Voice*, vol. 31, no. 2, p. 257–258, 2017, doi: https://doi.org/10.1016/j.jvoice.2016.06.001.

[79] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," arXiv, 2010.11929, 2021. [Online]. Available: https://arxiv.org/abs/2010.11929

[80] L. Beyer, X. Zhai, and A. Kolesnikov, "Better plain ViT baselines for ImageNet-1k," arXiv, 2205.01580, 2022. [Online]. Available: https://arxiv.org/abs/2205.01580

Statement of Authorship

I herewith assure that I wrote the present thesis independently, that the thesis has not been partially or fully submitted as graded academic work and that I have used no other means than the ones indicated. I have indicated all parts of the work in which sources are used according to their wording or to their meaning.

I am aware of the fact that violations of copyright can lead to injunctive relief and claims for damages of the author as well as a penalty by the law enforcement agency.

Magdeburg, September 13, 2024

_____

Signature